

Visual sensing technologies in robotic welding: Recent research developments and future interests

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ABSTRACT

Vision-based sensors play an essential role in robotic welding, which attract increasing attention by scholars because of its non-contact characteristic, high precision, fast detection, and strong adaptability. This paper summarizes the recent research developments of vision sensor-based intelligent robotic welding and analyzes their characteristics and application scenarios in the welding process. The applications of vision sensing technologies are illustrated, especially in visual calibration, initial welding position guidance, weld seam tracking, weld pool monitoring, correlation deformation detection, and defect identification. At the end of this paper, the promising development directions of intelligent robotic welding technologies are discussed, including the hardware developments of vision sensors, optimization of image processing algorithms, the combination of passive vision and active vision, multi-sensor information fusion, and applications of artificial intelligence.

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1. Introduction

Due to the high requirements for precision and efficiency, a large number of industrial robots were used in the welding industry. The welding robots can be divided into three generations [1]: the first generation of welding robots depends on the “teaching and playback” mode for training and working, which does not have sufficient adaptability to the changeable external environment. This traditional mode cannot automatically correct possible errors under non-ideal conditions, such as assembly error and thermal deformation [2,3]. The second generation of welding robots has sensors that act as eyes and can respond to changes in the environment. However, most of the complex operations still require human decisions. The third generation of welding robots not only has sensors as eyes, but also has a brain, which is capable of thinking and making decisions. By applying various algorithms, the robot can identify and reason the surrounding changes with independent decisions [4]. At present, however, most of the third-generation welding robots perform only partial automation. There is still a big gap to realize complete self-learning and empirical induction. Up to

now, the first two generations of welding robots still dominate the industrial market. In recent years, the emerging concept of industry 4.0 and the research in machine learning have pointed out a new direction for intelligent robots. Therefore, with the development of modern manufacturing technologies, research on intelligent welding robots has become an inevitable trend [5]. The development of industrial robots is summarized in Fig. 1.

Intelligent welding enables robots to imitate the operations of skilled welders in the welding process, including information acquisition, reasoning, decision-making, process control, and optimization [6]. A typical configuration of a robotic welding system is shown in Fig. 2. To realize automatic welding, the scholars need to solve critical problems in each link, such as initial welding position guidance, seam tracking, weld pool monitoring, penetration control, real-time deformation detection, and defects identification. The technical composition in the robotic welding system is shown in Fig. 3 [7]. Related technologies in each welding process were extensively studied in recent researches for applications of intelligent welding robots [8,9].

Sensors play an essential role in intelligent robotic welding. Using sensors to obtain information from the external environment is the prerequisite for solving those practical problems in the welding process. At present, robotic welding uses a variety of sensing methods, such as visual sensing [10,11], arc sensing [12,13], acoustic sensing [14], and vibration sensing [15]. Compared with other

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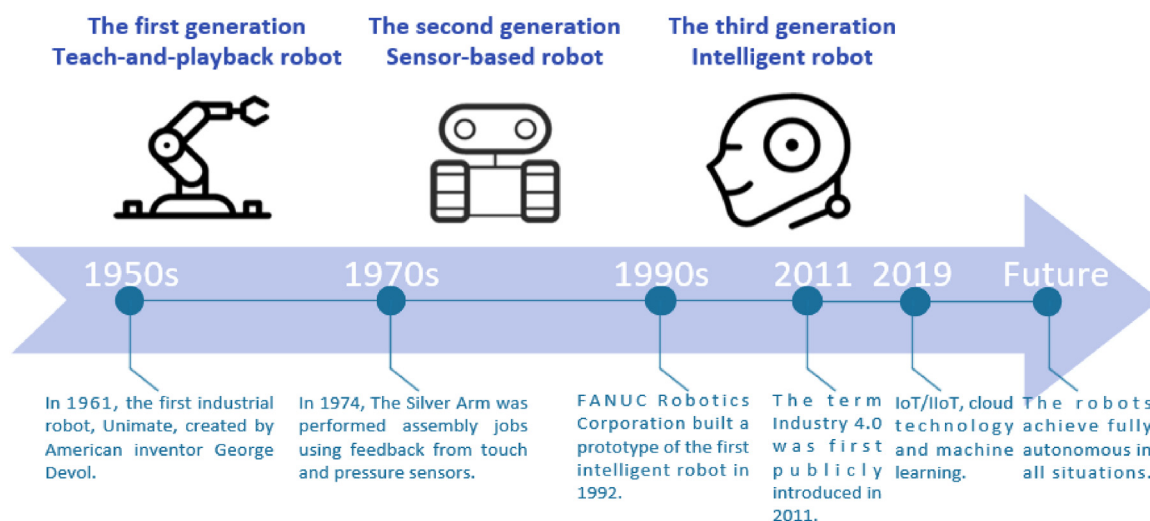


Fig. 1. The development of industrial robots.

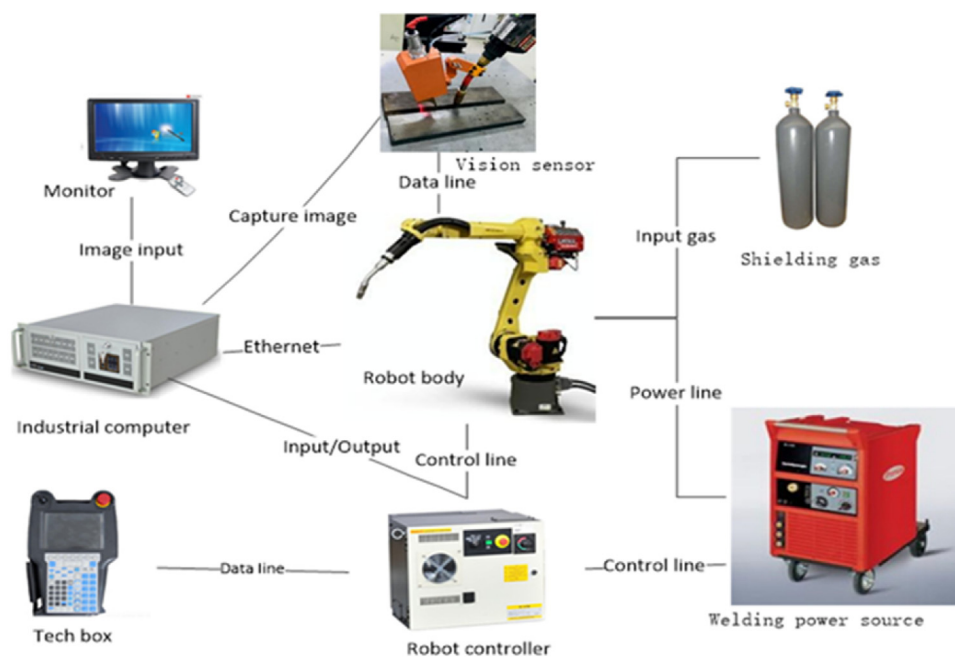


Fig. 2. The robotic welding system.

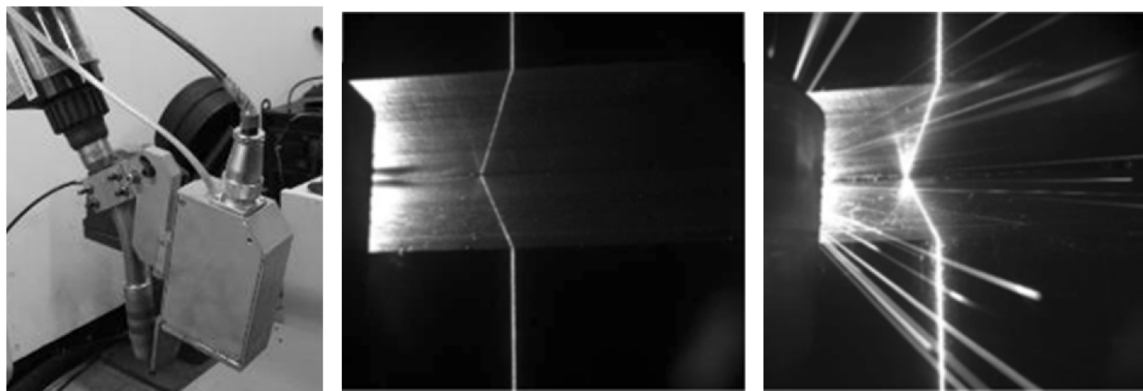


Fig. 3. The technical composition of intelligent robotic welding. Fig. 3 The active vision sensor and welding images [21].

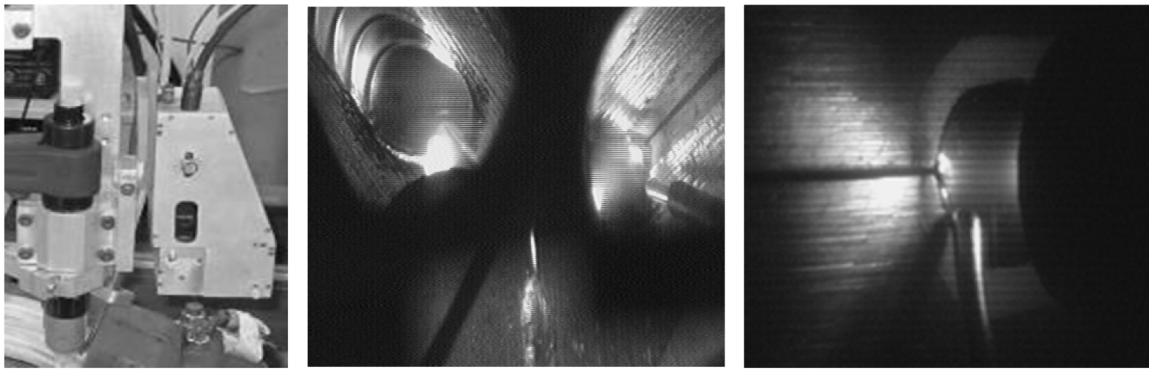


Fig. 4. The passive vision sensor and welding images [22].

Table 1
Comparison of passive vision sensor and active vision sensor.

Category	Active vision sensor	Passive vision sensor
Image processing	Simple	Complicate
Arc interference	Free from disturbance	Susceptible to disturbance
Joint type identification	Easy	Hard
Structure of sensors	Additional light source	No additional light source
Molten pool monitoring	Incapable	Capable
Application	Commercialized (Seam tracking)	Laboratory study (Penetration control)
Price	Expensive	Economical

sensing methods, vision sensors attract increasing attention with the advantages of non-contact characteristics, high precision, fast detection, and strong adaptability [16,17]. Even though for some specific seam types such as V-groove, the arc sensors can be more efficient than the visual sensor, other advantages of visual sensors still remain noteworthy.

The visual sensing technology can be divided into passive vision methods and active vision methods according to whether additional light sources are added. Passive vision directly collects the images of the welding area, which can obtain sufficient and more intuitive information. Without position error and time lag, this method can monitor the molten pool in real-time. However, strong noise interference, such as arc splash and smoke, brings great challenges to image processing [18]. Active vision often uses structured light or laser to project onto the surface of the workpiece and uses charge-coupled devices (CCD) cameras to obtain the information of the welding area indirectly. This method has a stronger anti-noise ability and can obtain characteristic parameters or information of joint type [19]. Active vision is widely used in robotic welding, but the advanced detection error and the deficiency of pool-monitoring also limit its application scenarios [20]. Typical welding images of passive vision and active vision are shown in Fig. 3 and Fig. 4. The comparison of passive vision and active vision is shown in Table 1.

For visual sensing technologies associated with robotic welding, the scholars have made much research as well as significant achievements. Yuming Zhang from the University of Kentucky and Shanben Chen from Shanghai Jiaotong University have conducted long-term research on vision-based robotic welding technology, and have also achieved fruitful results on different parts of the welding process [22–25]. Because of the critical role of visual sensing technologies in the robotic welding process, this paper summarizes the research and applications of visual sensing technologies, including calibration, initial welding point guidance, weld seam tracking, weld pool monitoring, deformation detection, and

defects identification, as shown in Fig. 5. This paper expounds on the characteristics and advantages of advanced visual sensing welding systems in each section. In the last part, promising development directions of visual sensing technologies are analyzed to provide some inspirations for the future research of vision-based robotic welding.

2. Visual calibration in robotic welding

Visual system calibration is a critical step in intelligent robotic welding, which directly affects the welding control accuracy in the subsequent processes, such as guiding, positioning, and seam tracking. There are three kinds of common calibration methods in robotic welding: camera calibration, hand-eye calibration, and structured light plane calibration. The imaging coordinate transformation matrix of camera calibration determines intrinsic parameters. At present, many camera calibration algorithms are widely used, such as the flexible technique for camera calibration from Zhang [26]. The hand-eye calibration is to determine the relative installation position between the camera and the welding torch. To simplify the calibration process and reduce cumulated error, many researchers conduct camera calibration and hand-eye calibration simultaneously. Besides, the calibration of the structured light plane, which describes the relative position of the laser plane under the camera, is also essential in active vision sensing systems. This paper will mainly discuss the vision-based hand-eye calibration and structured light plane calibration in robotic welding.

2.1. Hand-eye calibration

The robotic welding controls the weld parameters and the movement of the torch in real-time according to the data returned by vision sensors, so the hand-eye calibration is significant to achieve accurate control. Due to the relatively fixed mechanical structure of the robot, the relative position between the camera and the welding torch will not change significantly, so many traditional methods based on kinematic parameters have been studied and used in hand-eye calibration [27]. Daniilidis (1999) [28] utilized dual-quaternion parameterization to solve the hand-eye rotation and translation with singular value decomposition. Strobl et al. (2006) [29] proposed a metric of rigid transformations and corresponding error models for nonlinear optimization. Zhao (2011) [30] proposed a novel global linear convex optimization calibration algorithm. Carlson et al. (2015) [31] solved the rigid transformation matrix of non-parallel planes by a two-step iterative method.

In actual welding processes, however, some kinematic parameters of the robot will inevitably change slightly, such as joint deviation. These small errors will have a significant impact on positioning accuracy [32]. The vision-based calibration method can

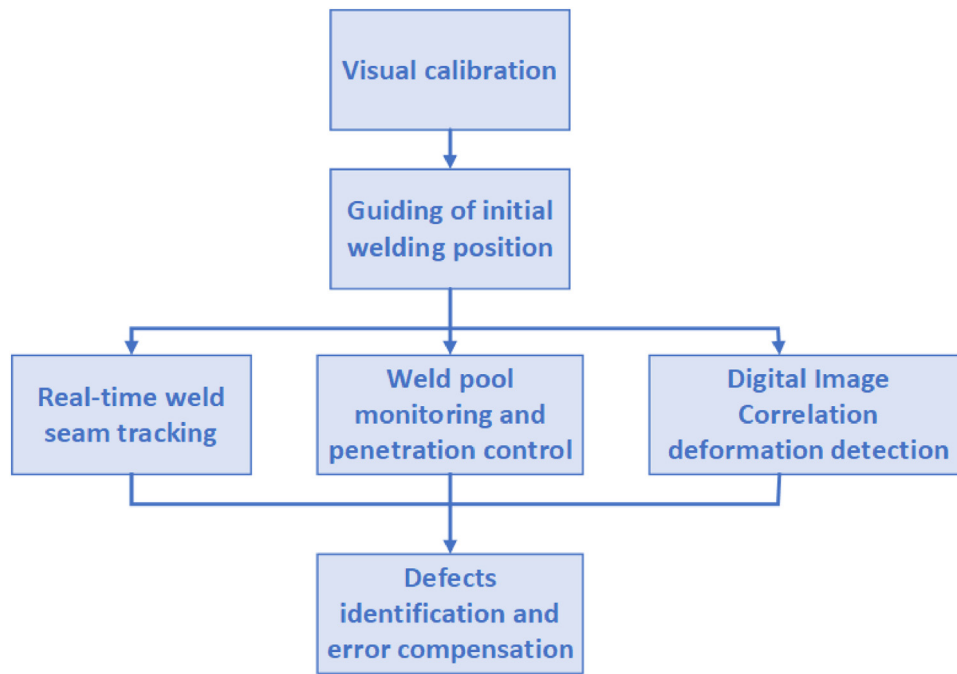


Fig. 5. The flow chart of vision-based technologies in robotic welding processes discussed in this paper.

solve these problems and meet the requirements of online operation. Also, vision sensor calibration does not need physical contact and precise measuring equipment. Because of the low-cost advantage and simple calibration processes, it is widely used in robotic welding systems.

At present, the parameter matrix of the hand-eye transformation can be solved by the traditional calibration method with a reference object or active vision-based calibration method.

2.1.1. Hand-eye calibration method with reference objects

The traditional hand-eye calibration method is to obtain images of reference objects by cameras in different locations and poses. By solving the relation equations between images and reference objects in different poses combining with corresponding attitude parameters, the hand-eye transformation matrix can be known.

Classic hand-eye calibration usually sets up a stereo vision system based on two CCD cameras or a single camera in multiple positions to obtain images from different perspectives. Due to the wide field of view, the measurement accuracy can be improved. Compared with monocular vision, stereo vision can directly measure the depth of objects from the image plane with high efficiency, so it is widely used in the hand-eye calibration system.

Tsai et al. (1989) [33] introduced a classic 3D robotics hand-eye calibration method with a CCD camera installed on a robot with seven degrees of freedom. Calibration was finished by homograph transformation and a glass calibration block with black discs. It could reach the accuracy of 12.516 mil translation error and 2.782 mrad rotation error with high speed. The accuracy of this method could be further improved by a more accurate 3D calibration object and additional measurement stations.

Malm et al. (2003) [34] proposed a calibration method of two cameras in a stereo configuration, which extended the plane-based intrinsic camera calibration to hand-eye calibration. The proposed method used a planar checkerboard calibration object and determined the hand-eye relationship merely depend on pure translational motions with translation direction or length. This system put small demands on the degrees of freedom of the robot vision system, which was much adaptive in practical applications.

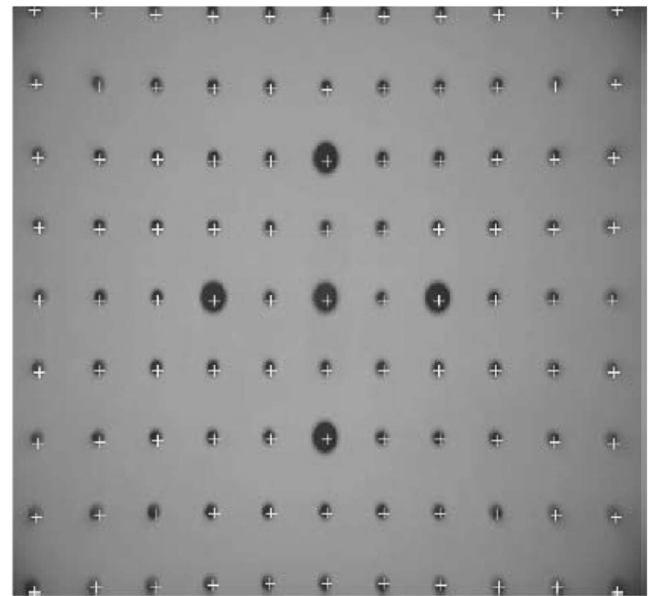


Fig. 6. Small black circles in the calibration pattern are used to deal with lens distortion; White cross-bars are superimposed to verify the accuracy [37].

Dinham et al. (2010) [35,36] presented a developed camera-based stereo vision and hand-eye calibration algorithm in a robotic arc welding system. With the non-linear optimization incorporating the robot kinematic model, the triangulation problem could be improved, and the absolute position errors could be reduced. An accuracy of 1 mm was achieved according to the experiment. Nevertheless, it can be noticed that the error tends to increase when the camera was moved further away from the workpiece.

To solve the problem of lens distortion, Motai et al. (2008) [37] presented a new model for hand-eye robotic calibration with multiple viewpoints based on monocular vision. A particular 3D calibration pattern with an array of black circles and white cross-bars was used, as shown in Fig. 6. This method integrated estimations of

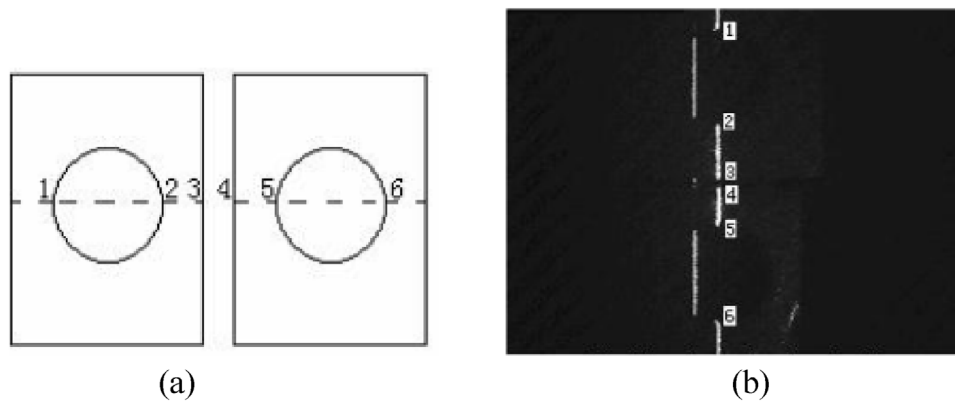


Fig. 7. (a) Calibration objects. (b) Image of laser projection on calibration objects [38].

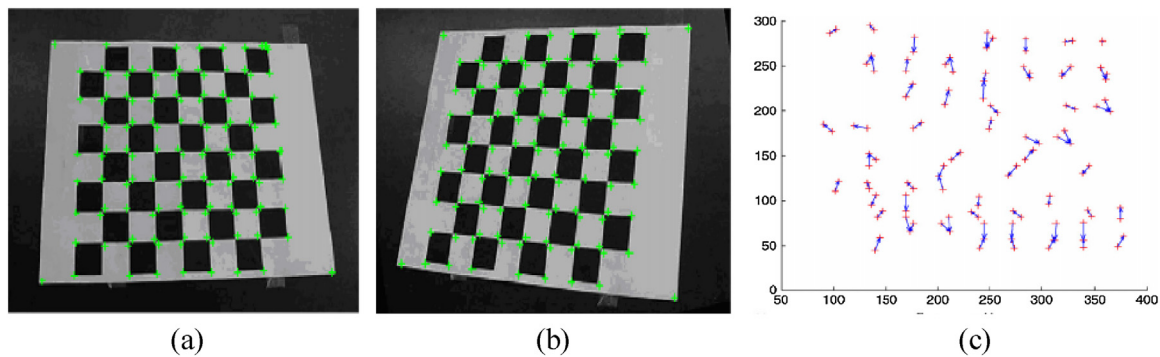


Fig. 8. (a) Corner detection of image 1. (b) Corner detection of image 2. (c) Feature matching [39].

camera parameters obtained by utilizing an extended Kalman filter. Snapshots of the calibration board were taken from five different active generated viewpoints to estimate the 3D coordinates with a final accuracy of 3 mm.

2.1.2. Active vision-based calibration

In the active vision-based calibration method, auxiliary light sources such as laser are introduced to project onto the surface of calibration targets. The hand-eye transformation matrix can be calculated by collecting the coordinates of feature points in captured images. Active vision-based calibration methods usually have high calibration accuracy, but it simultaneously needs high-cost external equipment.

Liu et al. (2008) [38] presented a laser vision robotic welding system with a single camera, which could simultaneously carry out camera calibration and hand-eye calibration. The laser was projected to two cubes, which have a hole in each along the dashed line. The calibration objects and laser images are shown in Fig. 7. The coordinates of feature points were extracted from the images, while the nonlinear least-square optimization determined camera parameters and the hand-eye matrix. The experiment proved that the proposed method could reach an accuracy of 0.12 mm.

2.1.3. Autonomous hand-eye calibration method

In recent years, autonomous hand-eye calibration systems without any manual intervention have been widely studied. By this kind of method, the robot can independently complete object detection, path planning, calibration parameters solving, and other online hand-eye calibration processes. Self-calibration is especially suitable for welding tasks in harsh environments such as high temperature or high pressure. Those tasks can be finished with satisfactory precision and efficiency.

Meng et al. (2007) [39] proposed a monocular vision self-calibration method without multiple accurate 3D feature points, which only referred to an exact calibration length in the coordinate system. Corner detection and correlation matching were applied to match images of different scenes, as shown in Fig. 8. The camera attitude could be estimated by factor decomposition, and the kinematics identification could be carried on accordingly. Accuracy of 0.369 mm could be reached, but it needed multiple measurements to reduce error.

2.1.4. Uncalibrated hand-eye system

In addition to the methods mentioned above, uncalibrated hand-eye systems are also proposed recently as alternatives when the traditional calibration methods are challenging to apply. This kind of system does not calculate camera parameters or robot kinematics model in advance and also does not need recalibrations in different environments, thus greatly simplifying the automatic welding process. However, uncalibrated systems are less accurate than calibrated systems in subsequent welding processes.

Yin et al. (2018) [40] proposed an uncalibrated seam tracking system for modular robots with low overall stiffness. In this method, the brightness of all pixels in the weld image was directly used as a visual feature. The feedback control was designed according to the difference between obtained images and expected images of the weld seam, as shown in Fig. 9. Experimental results showed that this method could achieve the accuracy of 1.06 mm in tracking tasks for lap weld seam and stitching weld seam.

2.2. Structured light plane calibration

Active vision systems based on structured light obtain information by projecting a laser stripe on the surface of workpieces. On account of its high precision and anti-interference ability, active

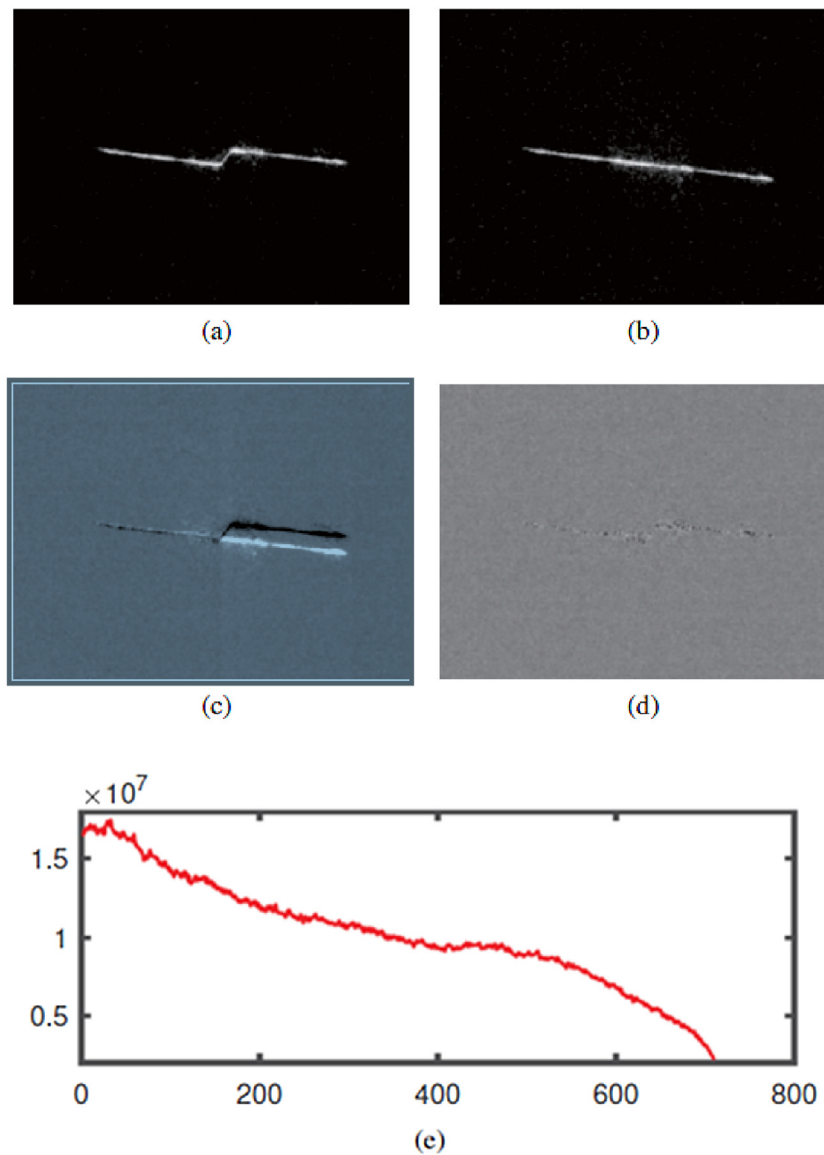


Fig. 9. Positioning tasks: (a) The desired image of lap weld seams. (b) Initial image of lap weld seams. (c) The difference between the two images at the initial pose. (d) The difference between the two images at the end of the motions. (e) The curve of absolute error [40].

vision calibration is widely used in vision-based robotic welding systems. The accuracy of the transformation between the structured light plane and coordinates of the CCD camera directly affects the precision of measurement. Therefore, it is imperative to calibrate the structured light plane to describe the relative position between the laser plane and the camera.

Similar to hand-eye calibration, there are two kinds of configurations when measuring three-dimensional information by structured light: stereo structured light and monocular structured light. Stereo structured light uses multiple cameras to obtain the images of the reference target at the same time and calculates the three-dimensional coordinates of the feature points. This method does not need to calibrate the parameters of the structured light plane.

Compared with binocular structured light vision, monocular structured light vision uses only one camera to obtain laser stripe images of reference objects. However, monocular vision cannot effectively obtain the depth of the reference objects, and the parameters of structured light planes need to be constrained. So, the structured light plane calibration is

essential for the precision of the monocular structured light vision.

In the process of structured light plane calibration, at least four noncollinear feature points on the laser stripe must be obtained. The traditional strained thread method [41], 3D Zigzag calibration object method [42], and other methods of obtaining calibration points often have complex processes. Those methods can only obtain a limited number of calibration points, which restricts the application scenarios. Some vision-based methods were developed to simplify this process.

Xu et al. (1995) [43] designed a particular trapezoidal and wedge calibration block, as shown in Fig. 10, by using the cross-ratio invariant method and calculation of the coordinates of multiple intersecting sample points on laser plane and calibration block. However, designing three-dimensional calibration blocks introduced additional complexity. The test results showed that the error along the length direction was less than 0.3 mm, and the error along the height direction was less than 0.2 mm.

To avoid designing unique reference objects, scholars researched on changeable planar calibration objects. Zhou et al.

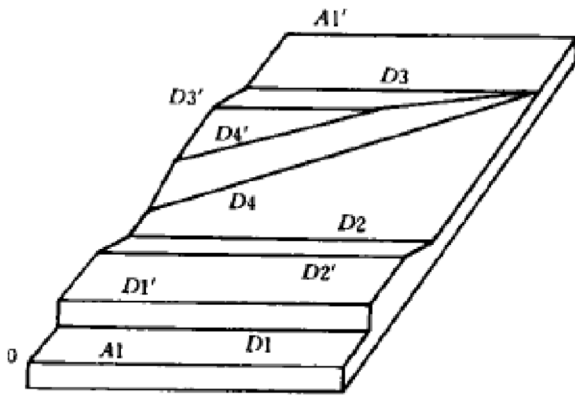


Fig. 10. A trapezoidal and wedge calibration block [43].

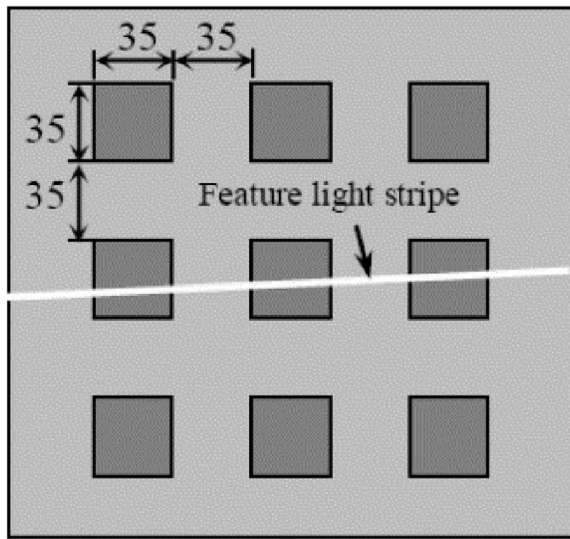


Fig. 11. The sketch of the planar calibration object with feature light stripes [44].

(2005) [44] employed the invariance of a cross-ratio to construct an arbitrary number of calibration points on the light stripe plane, as shown in Fig. 11. A planar calibration object with 3×3 square patterns was designed, which can move freely to obtain multiple calibration points and simplifies the calibration process. With at least two freely moved measurements, the coordinates of feature points can be calculated with a root-mean-square (RMS) error of 0.149 mm.

Fan et al. (2016) [45] projected laser stripes to a chessboard grid object, as shown in Fig. 12, to obtain a pair of feature points. The robot was then moved to a new position according to constraints to obtain another pair of non-collinear feature points. Structured light plane parameters were determined through these feature points. Without particular calibration objects and manual measurement, this calibration process is more straightforward. As a result, the root-mean-square (RMS) error is 0.078 mm.

Sun et al. (2009) [46] designed a universal calibration method for structured light sensors based on planar targets. The feature points of planar objects could be arranged irregularly, which reduced the difficulty of design and installation. A typical circular structured light was used to test the calibration performance. The quadric surface equation of structured light was fitted with the characteristic points of the light strip in five sections, as shown in Fig. 13. The results showed that this method was capable of calibrating structured light for any pattern, such as point laser, linear laser, and planar laser. The root-mean-square (RMS) error was 0.03 mm.

2.3. Summary

In the intelligent robotic welding, visual sensing technology plays an important role in welding guidance, seam tracking and welding quality monitoring, and the calibration of vision sensors is the fundament of welding systems. The control accuracy is directly determined by the final calibration precision of visual systems.

Visual calibration of welding systems is a complicated and tedious process, which has high requirements on precision and robustness. How to calibrate quickly, stably, and accurately is always a great challenge. Many research works have been done by scholars, and a series of research results were achieved, such as a simplified calibration process, automatic calibration, non-calibration system, self-applicable calibration methods. Especially for the more complicated calibration process in laser vision systems, this paper summarizes the research achievements of hand-eye calibration and laser plane calibration respectively.

3. Guiding of initial welding position

In the robotic welding process, the first step is to guide the welding torch to the initial welding point. The traditional "teaching and playback" method needs human operations and is time-consuming. Therefore, the automatic positioning of the initial welding points becomes a critical problem in intelligent robotic welding. The vision-based positioning method is widely used because of its advantages of non-contact operation and high precision. According to whether additional auxiliary lights are added, the initial positioning guiding method can be divided into active vision method and passive vision method.

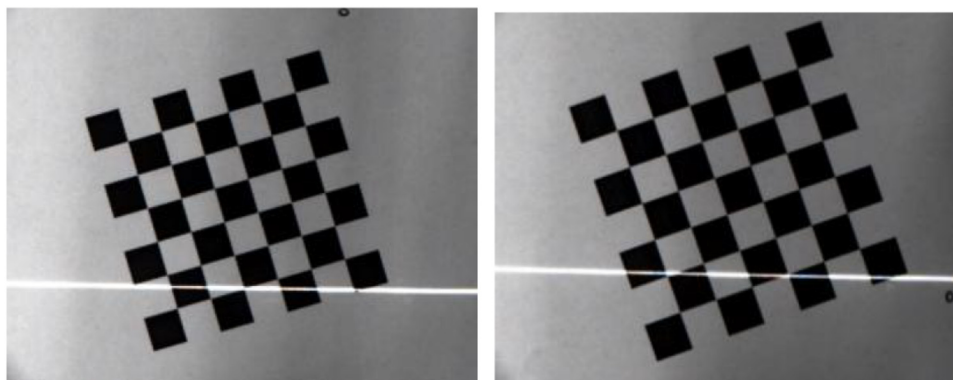


Fig. 12. Two sets of chessboard images with non-collinear laser stripes [45].

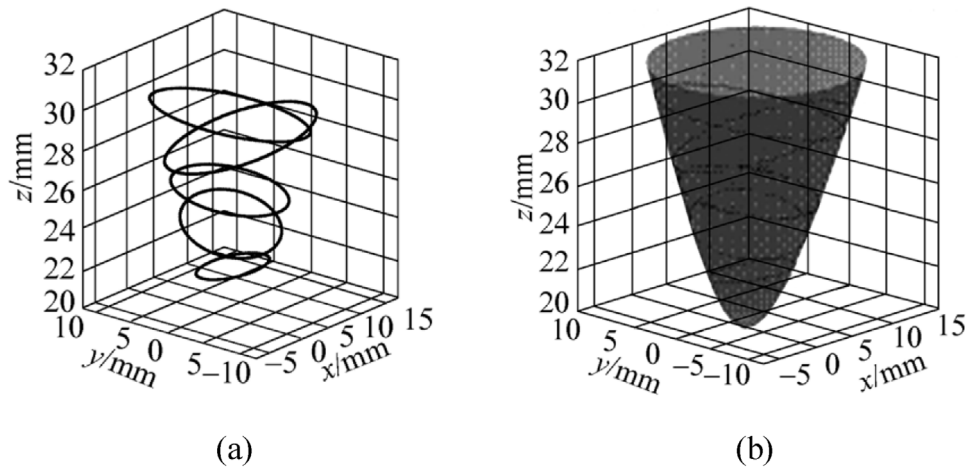


Fig. 13. (a) Circular structured light in five sections. (b) The quadric surface of structured light [46].

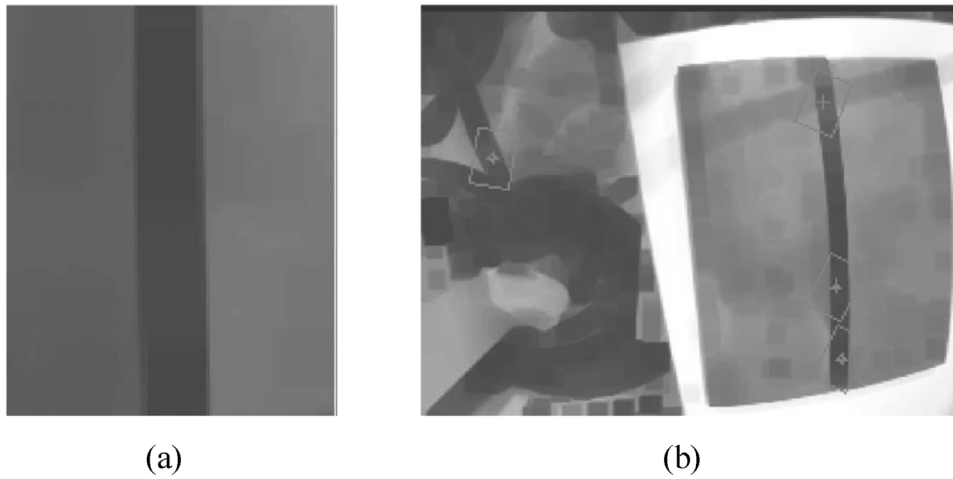


Fig. 14. (a) Template image. (b) Global matching image [47].

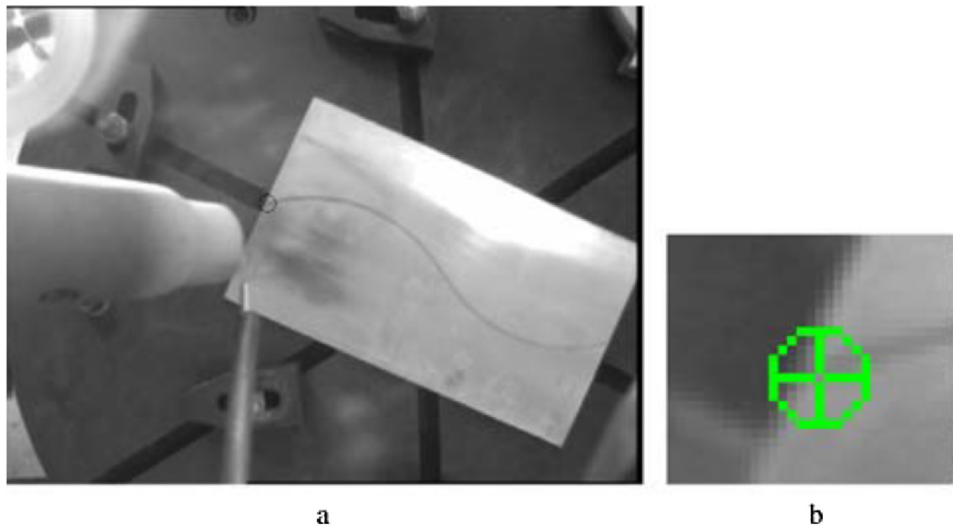


Fig. 15. (a) Image after pre-processing. (b) The template after amplification [48].

3.1. Passive vision-based guiding method

The passive vision-based initial position guiding method is to use a CCD camera to directly collect the image of the workpiece,

and identify the initial position with specific characteristics by image processing. This method is convenient, which does not need any additional device. Because the precision of image recognition determines the accuracy of the passive vision-based initial point

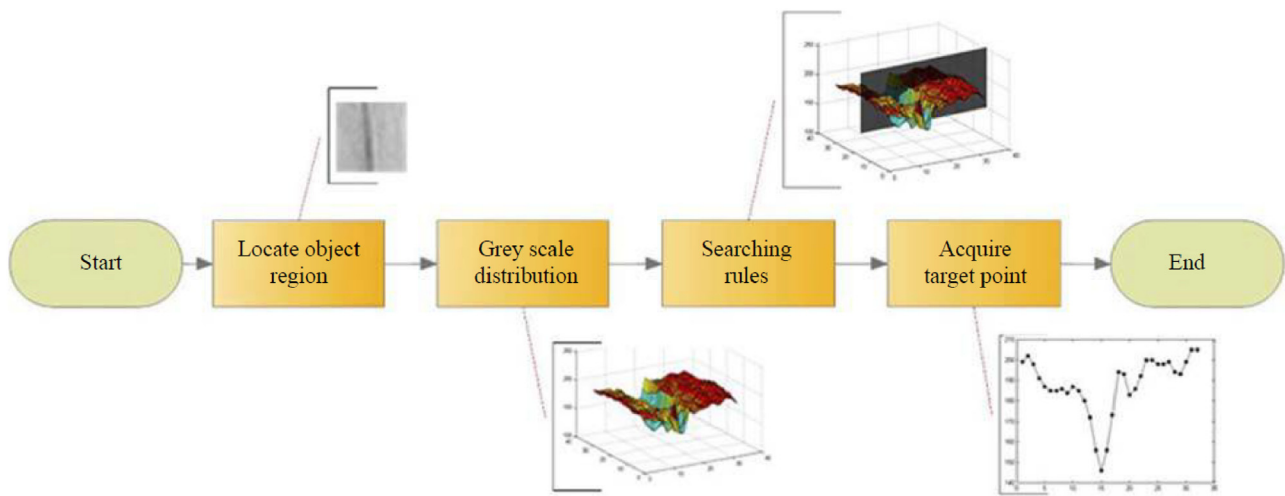
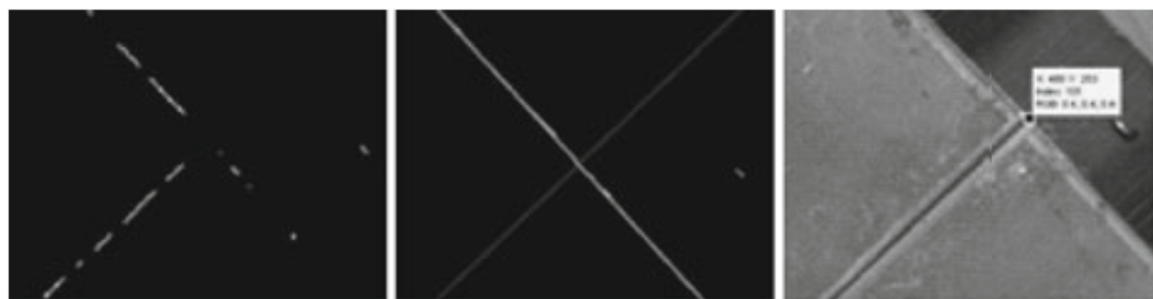


Fig. 16. The process and effect of the greyscale scanning method (GSCM) [49].

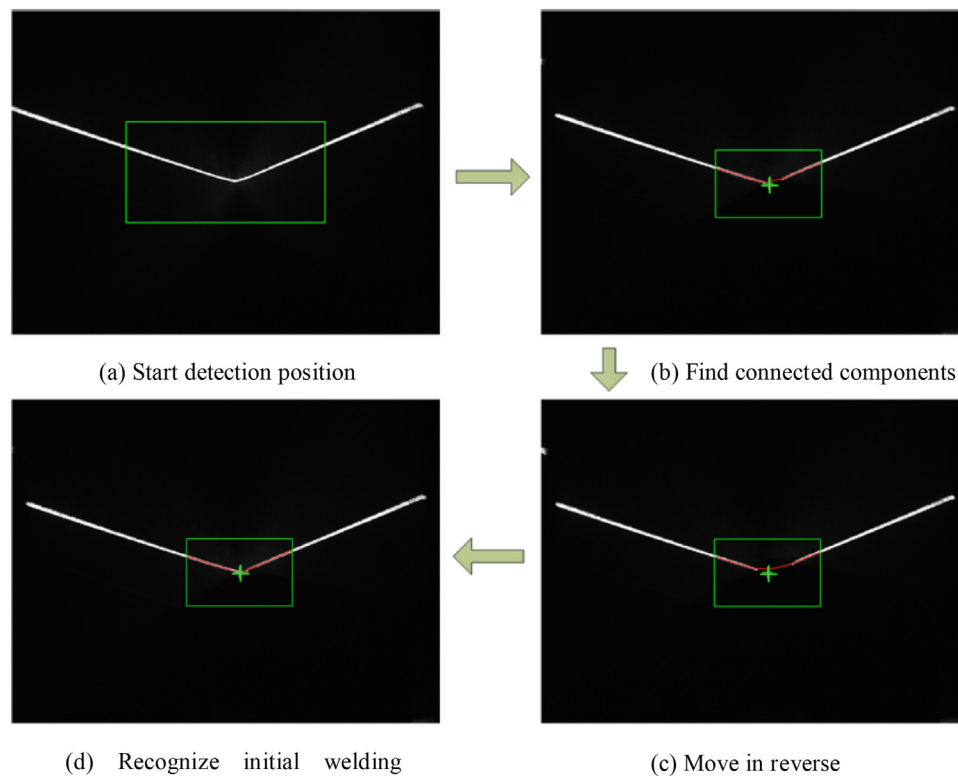


(a) Image erosion

(b) Hough transform

(c) Feature extraction result

Fig. 17. The procedure of feature extraction [50].



(a) Start detection position

(b) Find connected components

(d) Recognize initial welding

(c) Move in reverse

Fig. 18. The process of locating an initial welding point [53].

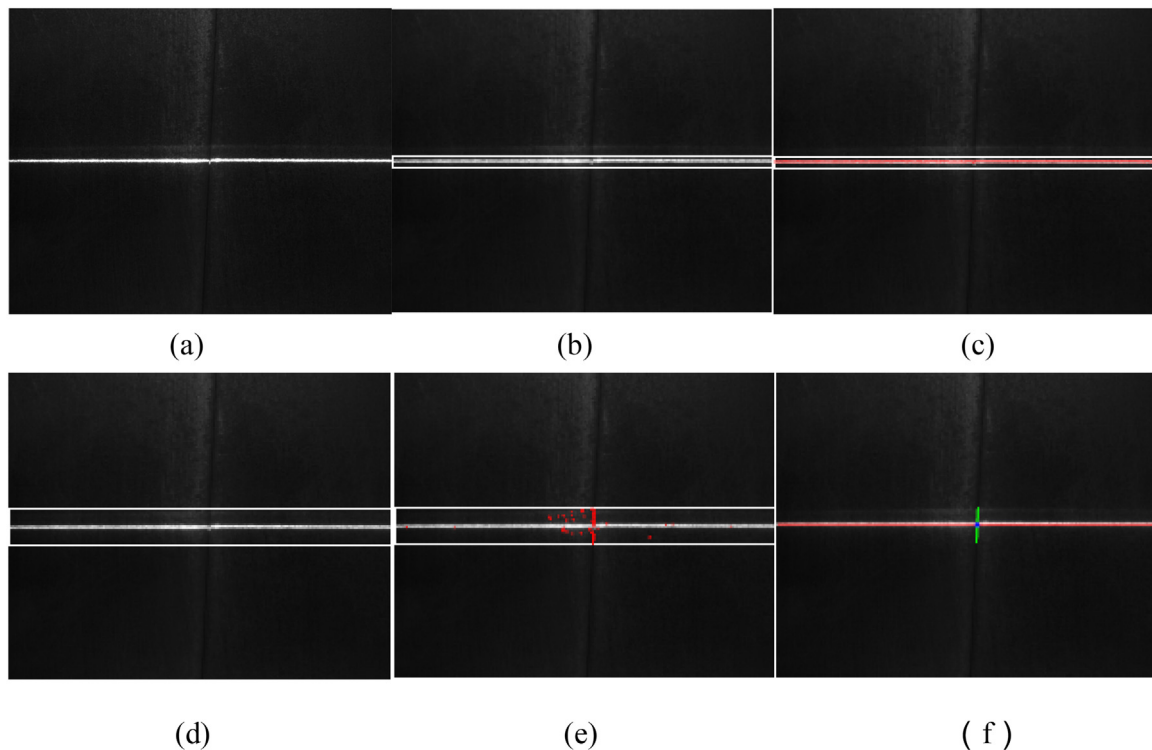


Fig. 19. The image processing procedures. (a) The image after preprocessing. (b) ROI of laser stripe. (c) Center profile of laser stripe. (d) ROI of the weld seam. (e) Center profile of weld seam. (f) The laser stripe, weld seam, and feature point [54].

guiding method, it is essential to choose appropriate image processing methods to identify feature points.

Zhu et al. (2005) [47] applied a two-step template matching in recognition of the initial welding point of linear weld seam flat butt joint. It utilized the correlation between brightness patterns in the local pixel neighborhood in one image and the corresponding area in the standard template image. After the approximate location of the weld seam was found, as shown in Fig. 14, a method of successive approximation was applied to determine the exact initial point, which was defined as the point with the least value of horizontal coordinate. This method is easy to implement, but its accuracy highly depends on accurate templates and clear images of the weld seam.

Chen et al. (2006) [48] extended the template matching method, as shown in Fig. 15, to curved weld seam butt joint in the planar welding process. This method was specially designed for aluminum workpieces, whose strong reflectivity introduced additional noise in image processing. Polynomial interpolation technology of pixel was applied after template matching to subdivide the pixel coordinates for high precision. After the transformation of coordinates, the initial point of the weld seam could be accurately localized with a 3 mm average position error. The constraint conditions were given according to the neighborhood pixel values, which is relatively simple and may not achieve the desired effect in complex scenarios.

In addition to template matching methods that require manual-selected templates, Wei et al. (2010) [49] proposed an autonomous initial point guiding method with the configuration of a single camera in different positions. The processing of the acquired image, including smoothing, threshold segmentation, and edge detection, was implemented to localize the weld seam. An improved Harris algorithm was proposed to extract corners of the flat butt joint. The greyscale scanning method (GSCM), as shown in Fig. 16, was used for those workpieces with specific marks. According to the experiment on both flat butt joints and flat flanges, the max abso-

lute error of straight seam and the curved seam was less than 2.3 mm.

Wei et al. (2012) [50] applied image morphology in the initial point detecting. The edge of the acquired image was detected by the morphology method, and then multi-directional linear structuring elements were selected by the direction filter. By the adaptive Hough transformation, the initial welding position can be obtained as the intersection. The procedure of image processing is shown in Fig. 17. As a result, the distance deviation is no more than 5 pixels in real-time image processing of straight seams.

Chen et al. (2010) [51] designed a stereo vision system with two CCD cameras to find the initial welding point with a two-step coarse-to-fine method. The first step was to solve the curve functions of seam and workpieces boundaries by fitting, and the intersection point is regarded as the initial position. The second step was to establish a small window and obtained exact coordinate by corner detection. The results showed that the method could get the position of the initial point with an accuracy of 1.1 mm for both regular and irregular weld seams.

3.2. Active vision-based guiding method

The active vision-based initial point guiding method utilizes additional light sources such as structured light to assist in locating feature points. Because the laser stripe is not easily disturbed by external noise, the active vision method has better robustness, adaptability, as well as a higher accuracy. Also, the active vision positioning method is suitable for T-type joint, lap joint, and other complex joint types, which are difficult to be detected by passive vision methods.

Wang et al. (2017) [53] proposed a method to identify the initial welding position of fillet weld seams. Laser scanning was utilized to capture mutations of laser stripes at the initial welding position in continuous frames. After two connected components were found, the robot would move back and forth slowly at the end of the weld

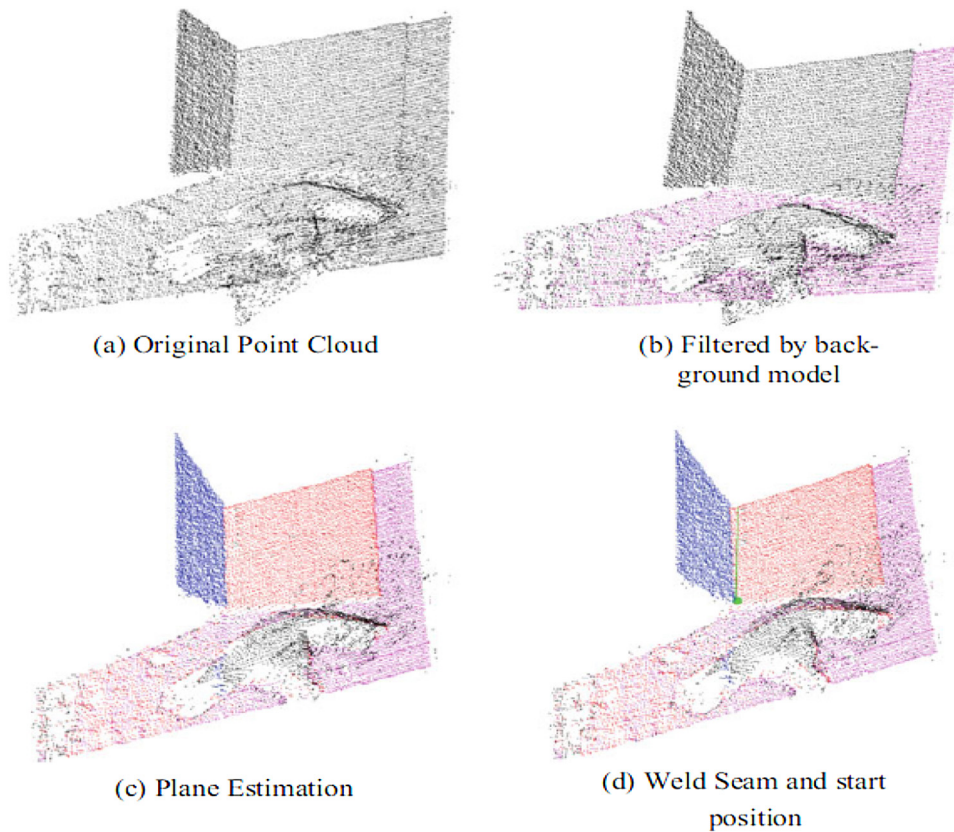


Fig. 20. Fitting and identification in a noisy environment [55].

seam to precisely locate the initial welding point. The procedure of image processing is shown in Fig.18. The absolute error was less than 0.8 mm.

To find the initial welding position of curved micro weld seams, Fan et al. (2019) [54] designed a structured light vision sensing system including a narrow-band filter and light-emitting diode (LED) auxiliary light. After obtaining clear laser stripe images, the coordinates of the initial welding points were calculated by the gray matter center method and random sample consensus (RANSAC) line extraction. The procedure of image processing is shown in Fig.19. As a result, the root-mean-square (RMS) error was less than 0.3 mm. By extracting corresponding features, the proposed method was also applicable for other joint types such as lap joint, fillet joint, and V groove joint.

Considering the inadequacy of traditional stereo vision positioning methods, which requires extra equipment, Zhang et al. (2018) [55] proposed a new initial point positioning method by using three-dimensional point cloud reconstruction as shown in Fig.20. After the center of laser stripes was accurately extracted from sampled video frames, the point cloud in the image was segmented by KD-Tree based background modeling and random sample consensus (RANSAC) fitting to fit the correct reconstruction model. The position of initial points could be found along the intersection of the fitting plane. Experiments showed that this method has high accuracy and good anti-interference ability, specifically for the weld seams with complex structures.

3.3. Summary

The guidance of initial welding points is an important research topic in autonomous robotic welding. The vision-based initial welding point positioning method can accurately determine the starting

position of each welding process without manual intervention, which includes active vision methods and passive vision methods.

Passive vision and active vision have their advantages and disadvantages for initial welding position guidance. Passive vision can be used in large welding scenes and can obtain rich information from a wide field of view. Of course, the complicated processing of passive vision is more vulnerable to external interference, and its accuracy and stability are not as good as active vision. However, the active vision can only obtain a small amount of information when the camera is partly blocked by three-dimensional workpieces. Its application is limited to the local field of view. At present, the more suitable way is to use passive vision for a wide range of rough guidance, and then to carry out precise guidance in local position, which combines two methods to complement each other.

4. Weld seam tracking

Weld seam tracking is a crucial problem in autonomous robotic welding. The robots are expected to automatically plan and control the movement of the torch by analyzing the image obtained by sensors. Automatic seam tracking replaces the traditional “teaching and playback” mode without the requirement of manual path planings in advance. The application of vision sensors in weld seam tracking can be divided into two modes: active vision and passive vision. Each mode has its advantages in different application scenarios. The summary of papers studied for vision-based seam tracking is listed in Table 2.

4.1. Passive vision-based seam tracking

Passive vision adopts a relatively economical and straightforward sensing system, which uses CCD cameras to directly collect images of weld seams without any auxiliary light source. By direct

Table 2
Papers of vision-based seam tracking methods.

Year	Authors	Visual sensing technology	Methodology	Type of welding and weld joints	Type and width of weld seam	Type and thickness of workpiece	Tracking accuracy
2013	Ye et al.	Passive vision	Two-window method, Sobel edge detection, Fuzzy PID controller	Pulse-MAG/ Butt joint	Straight and curved seam	Low carbon steel, 5 mm	Straight 0.5 mm, Curved 1 mm
2013	Nele et al.	Passive vision	Template matching with ROI	GMAW/ Square butt joint	Straight seam, 1.5 mm	Iron steel 510	0.1 mm
2014	Xu et al.	Passive vision	Improved Canny edge detection	GMAW/ Butt joint	Straight seam	Q235 steel, 2.5 mm	0.3 mm
2014	Chen et al.	Passive vision	Statistical filtering noise suppression, Hough transform by pixel gray distribution	CO ₂ gas shielded arc welding/ Narrow butt joint	Straight seam, Less than 0.2 mm	High-strength sheet steel	0.3 mm–0.4 mm
2015	Liu et al.	Passive vision	RANSAC and Kalman filter seam detection, Active contour driven by Bayesian classifier	MAG / V-groove butt joint	Straight seam, Rather wide and not constant (need torch oscillation)	Steel	–
2017	Li et al.	Active vision	Least-squares fitting, Junction identification by character string method	Butt joint (V-groove and angle welding)	Straight seam	Carbon steels, 10mm	–
2019	Du et al.	Active vision	Fast image segmentation with strong noise, CNN feature recognition	GMAW/ V-groove butt joint	Straight seam	Q235 steel, 20mm	0.26 mm–0.29 mm
2017	Fan et al.	Active vision	SVM-based modeling, BPNN weld seam type recognition	Butt joint(V-type and I-type) and lap joint	Straight and curved seam	–	–
2017	Zeng et al.	Active vision	Information fusion of 2D pixel coordinates and 3D point cloud data	3D narrow butt joint	3D curved seam, Less than 0.1 mm	Aluminum alloy, stainless steel, etc. 5 mm	0.24 mm
2016	He et al.	Active vision	Visual attention guidance model, Feature point extraction	MAG/ V-groove butt joint	Multi-layer and multi-pass weld seam (Weave welding)	Thick-plate-steel, 30mm	–
2018	Zeng et al.	Active vision	3D positioning by fusion information of directional and structured light images	MIG or MAG/ V-groove butt joint	Multi-layer and multi-pass weld seam	Low carbon thick steel	0.6 mm
2008	Xu et al.	Active vision, Circular structured light	Two-step segmentation technique, 3D position calculation algorithm	Arc welding/ Butt joint(V-type and I-type) and ramp joint	Straight and curved seam	–	1 mm
2016	Kiddee et al.	Active vision, Cross structured light	Template matching with ROI, Adapted line fitting algorithm	V-groove butt joint	Straight V-groove seam, 15.5 mm	–	0.2 mm
2017	Zhang et al.	Active vision, Grid structured light	Grid intersection extraction algorithm	GMAW	Multi-layer and multi-pass weld seam	Thick plate	0.5 mm
2018	Zou et al.	Active vision, Multi-line structured light	Gaussian kernel-based feature tracking, Adaptive fuzzy controller	Short-circuit MAG/ Butt, lap, and complex joint	Curved seam	–	0.32 mm

monitoring of arc, passive vision can obtain sufficient information from welding images, which is beneficial to the control system in the welding process. It is also an excellent complement to the active vision, which makes up the deficiency in weld pool dynamic monitoring. Many scholars have done researches on the weld seam tracking by passive vision [56].

In terms of image processing, Ye et al. (2013) [57] utilized the spectrum analysis and a dimmer glass to reduce most of the exposure light in the pulse metal active gas (MAG) welding process. They indicated that the fixed window method is not suitable for slight distance variation caused by thermal deformation between the torch and the working space. Therefore, they proposed a two-window method, as shown in Fig. 21, to extract the center of the seam and the weld pool separately. Edge detection by a Sobel operator was used in seam extraction, which is shown in Fig. 22. The precision of 0.5 mm for straight seams and 1 mm for curved seams could be proved by the test of fuzzy proportional-integral-derivative (PID) controller, but its adaptability and real-time performance need to be improved in complex environments.

Dilation. (e) Erosion. (f) Area filter. (g) Seam-like edges search. (h) Seam edges identification. (i) seam representation. [57]

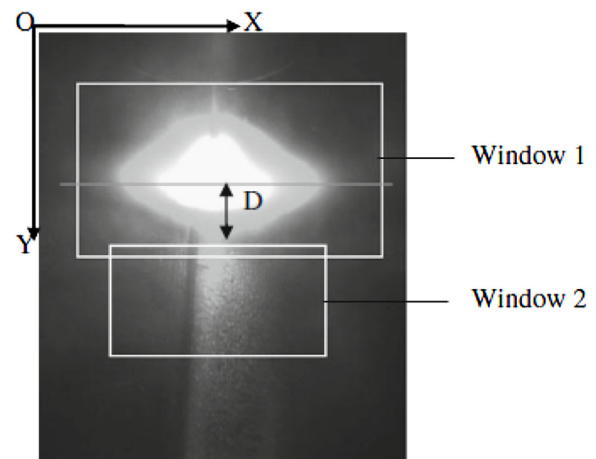


Fig. 21. The welding image with two analysis windows [57].

To improve the accuracy of contour extraction, Xu et al. (2014) [58,59] proposed a targeted approach to measure the deviation between the welding wire and the weld seam centerline in robotic GMAW. The procedures contained image preprocessing,

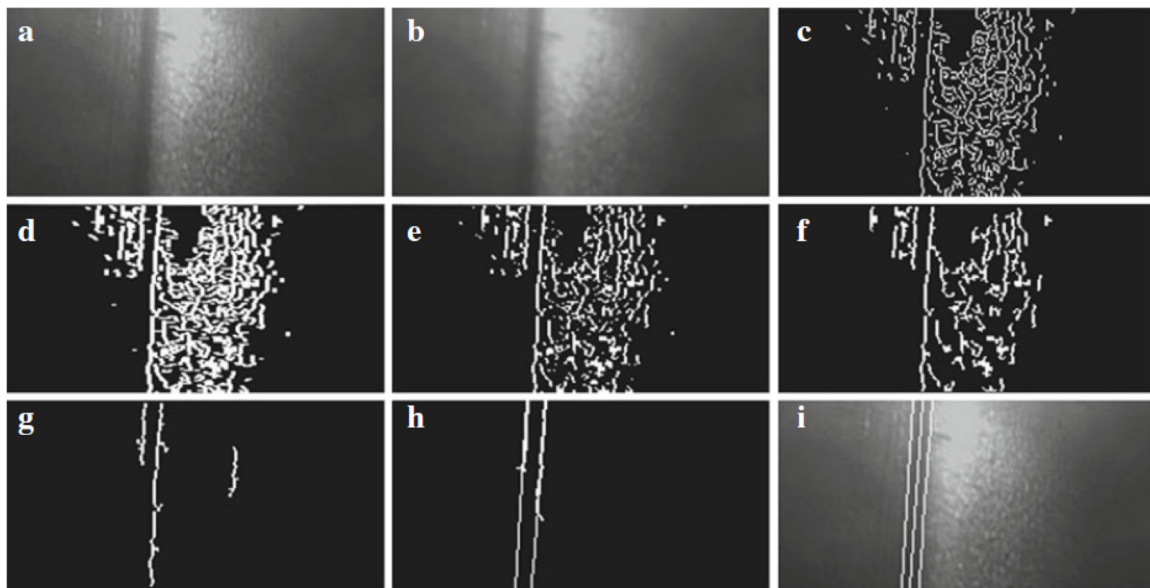


Fig. 22. Image processing of Window 2: (a) Original image. (b) Mean filter. (c) Edge detection. (d).

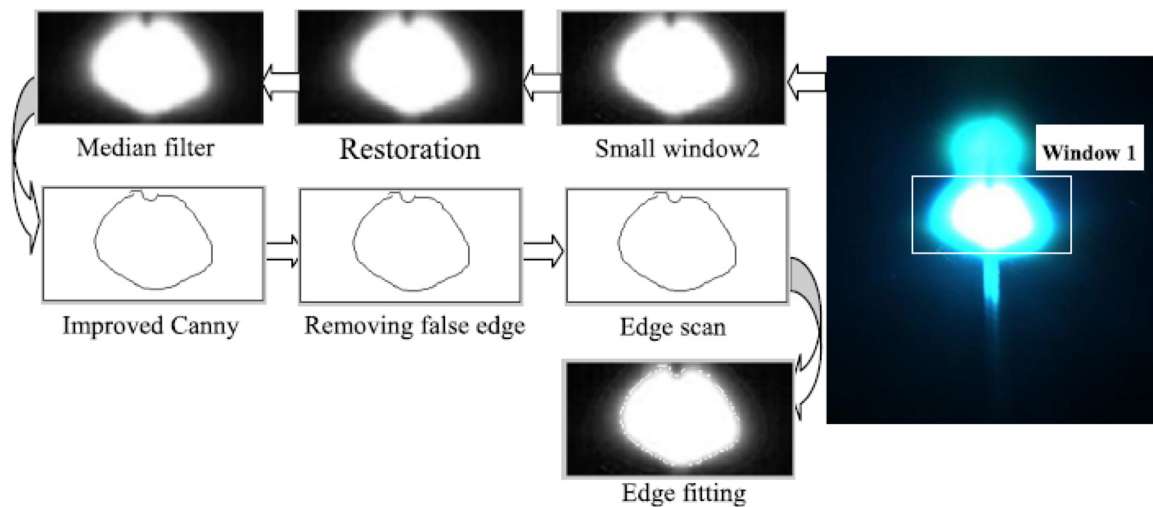


Fig. 23. The procedures of weld pool image processing [58].

edge detection by improved Canny operator, false edge removal, edge scanning, and characteristic parameters extraction, which is shown in Fig.23. A range of ± 0.3 mm could be achieved in experimental verification.

To improve the real-time performance of image processing algorithms, Nele et al. (2013) [60] designed GMAW control units and an image acquisition system by template matching. Three regions of interest (ROI), as shown in Fig.24, were defined to realize torch positioning and real-time seam tracking for square butt joint in a LabVIEW environment. The templates are shown in Fig.25. By matching the templates for each ROI, weld positions under different conditions (lighting and weld widths) can be quickly identified with an accuracy of 0.1 mm. An elaborately designed template and more accurate matching rules can further improve the precision of seam tracking.

For narrow seams, spatter and light interference is usually strong, which makes narrow gaps extremely hard to be detected. Chen et al. (2014) [61] proposed a statistic-based filter method for a double head welding robot to reduce the noise caused by various disturbances effectively. All points that possibly belong to the weld seam were screened out through the difference of pixel dis-

tribution, which is shown in Fig.26. Then least-square fitting and Hough transform were used to extract the characteristic lines of narrow seams. Seam tracking and initial point positioning can be done simultaneously, and however, it is noted that the tracking accuracy of 0.3 mm for narrow seams still needs improvement.

In order to get information about the weld seam and the molten pool, Liu et al. (2015) [62] captured frames for seam tracking and weld pool sensing respectively by changing the exposure time. This method is designed for V-groove joints with wide seams that required torch oscillations. Seam detection algorithms based on RANSAC and a Bayesian classifier were used to extract contours of the weld seams and molten pool, which are shown in Fig.27. The experiments on V-groove weld seams verified the feasibility of the proposed system.

4.2. Active vision-based seam tracking

Active vision is widely used in commercial applications, among which the most commonly used is laser-based structured light. It is mainly used for initial welding point guidance, weld seam iden-

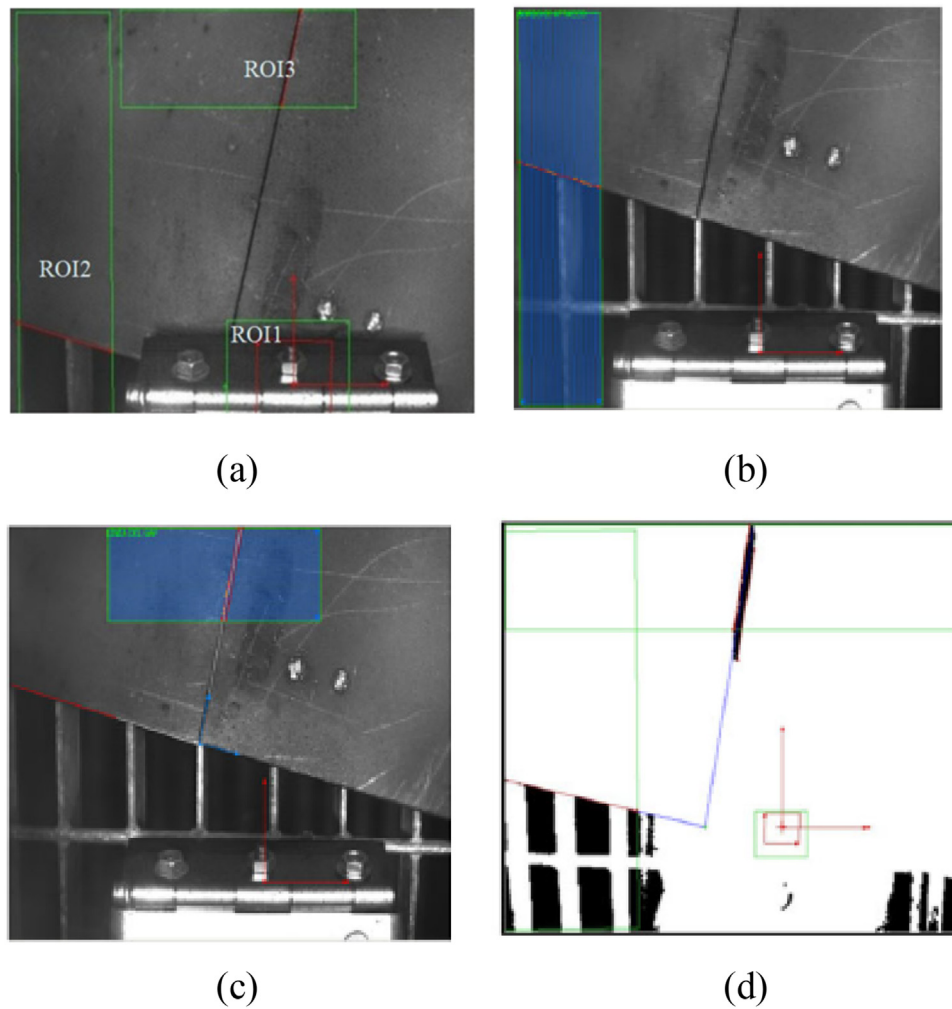


Fig. 24. (a) A image of a workpiece with three rectangular ROIs. (b) ROI2 indicates the lower edge of the workpiece with red lines. (c) ROI3 indicates seam gap edges with the red lines. (d) The filtered picture verifies the position of initial welding points [60].

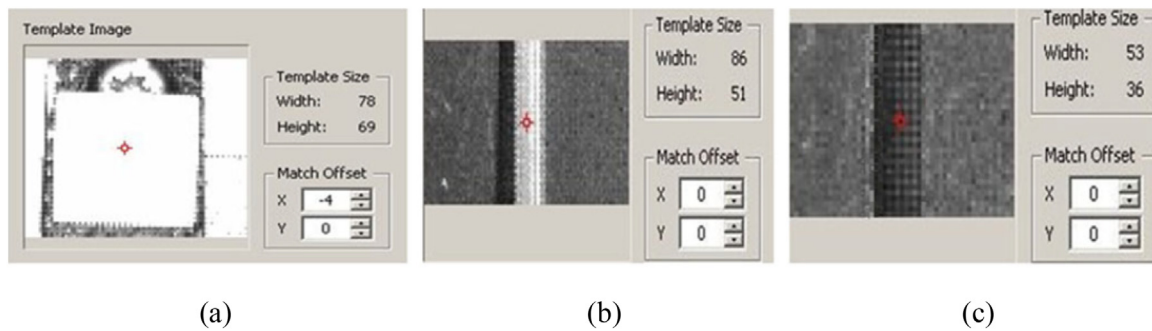


Fig. 25. (a) Torch template. (b) White-gap template. (c) Black-gap template [60].

tification and seam tracking. In particular, many researchers have made a series of remarkable achievements in welding seam tracking tasks. At present, many researchers have studied the weld seam tracking system with structured light methods.

4.2.1. Straight weld seam tracking

The straight weld seam is the most basic seam type, and the tracking process is more straightforward because of its regular shape. Therefore, the research of straight weld seams mainly focuses on noise removing and laser stripe identification. In actual welding processes, the shapes of laser stripes projected on the

workpieces are quite different due to diverse connection modes and groove shapes, such as butt joint, corner joint, and lap joint. A specific image processing algorithm will be applied to different types of weld seams. Thus, one of the significant tasks is to determine the position of feature points and identify the seam types [63].

Li et al. (2017) [64] researched image processing algorithms to identify the straight weld seam from a noisy environment and possible thermal deformation. The least-squares method was utilized to extract the profile weld seams. To identify the type of weld seams, character strings, as shown in Fig.28, were used to qualitatively

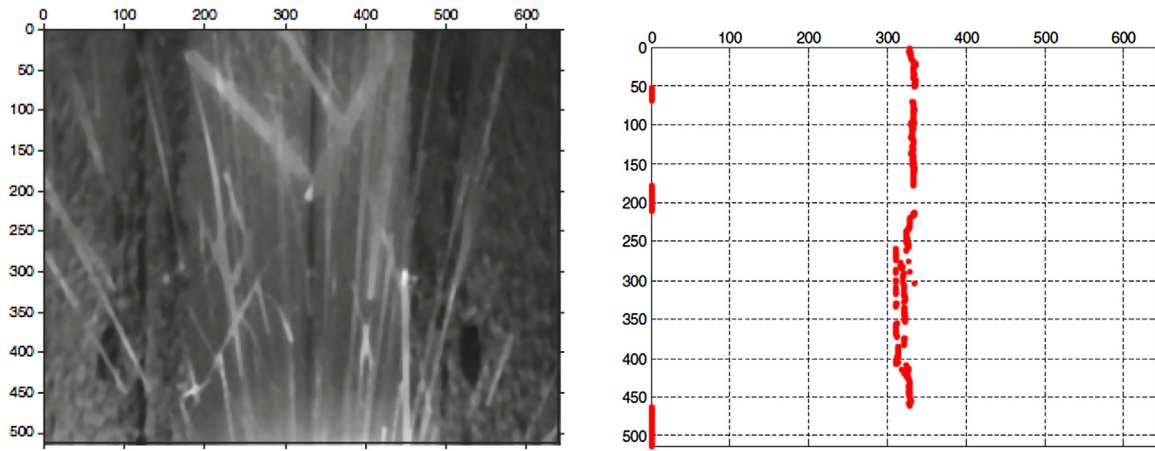


Fig. 26. The pixel coordinates of seam profiles extracted from welding images [61].

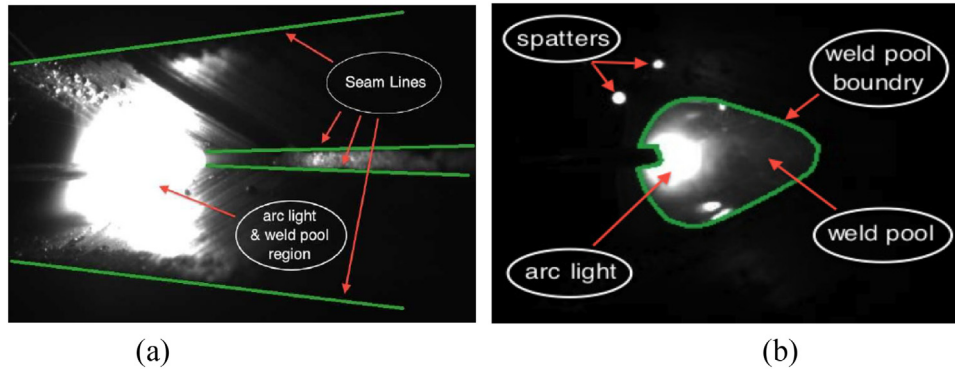


Fig. 27. (a) A typical frame captured for seam detection. Four lines that describe the V-Groove workpiece are marked in green. (b) A typical frame for weld pool monitoring. The boundary of the weld pool is manually annotated here and marked in green [62].

Qualitative description	Element	Symbol
Horizontal		h
Vertical		v
Upswept		u
Declivitous		d

(a)

Qualitative description	Relationship	Symbol
Connection without the angle		c_1
Connection with the anticlockwise angle		c_2
Connection with the clockwise angle		c_3
Horizontal break		g_h
Vertical break		g_v
Upswept break		g_u
Declivitous break		g_d

(b)

Fig. 28. (a) The definition of linear segment elements. (b) The definition of junction relationship elements [64].

describe the profiles, which might consist of different segments and junction relationships. Experiment on V-groove and fillet weld seams proved this method had high precision and stability, but its real-time performance and applicability on curved seams were not satisfactory.

To solve the processing problems of images with strong noise, Du et al. (2019) [21] utilized fast image segmentation, feature recognition by the convolutional neural network (CNN), and the feature search technique to identify the weld seam features accurately. The processing of noisy images is shown in Fig. 29. Automatic threshold selection reduced the difficulty of parameter adjustment and increased the stability of seam tracking. This method

performed an accuracy of 0.26 mm–0.29 mm in tracking and recognition tasks of noisy welding images.

Hou et al. (2020) [65] studied a teaching-free welding method based on a laser visual sensing system (LVSS) for robotic gas metal arc welding (GMAW). After a fast-unified calibration method, an image processing method based on prior knowledge were used to obtain feature points with sub-pixel accuracy in real-time. The flow chart of image processing of the improved Steger method is shown in Fig. 30. The online welding trajectory planning method proved the feasibility of the proposed welding teaching-free system on V-grooves and fillet welds.

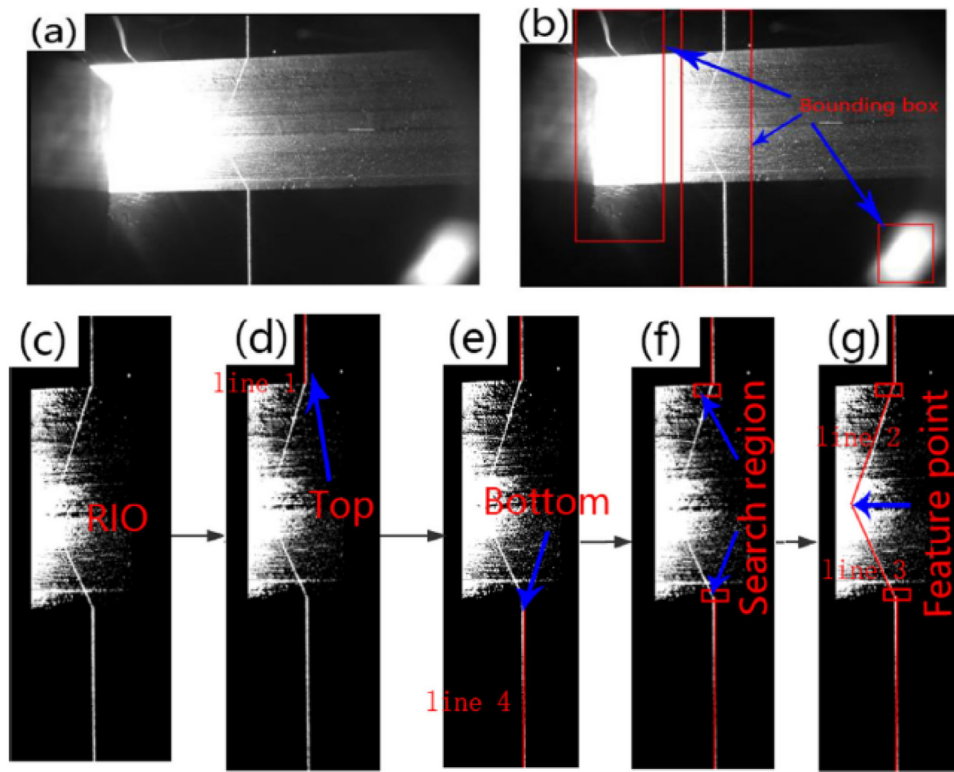


Fig. 29. The processing of images with strong arc noise. (a) Original image. (b) Divided image. (c) Using CNN model identification ROI. (d) The upper end of the search. (e) The lower end of the search. (f) The search of corner regions. (g) Obtained feature points [21].

4.2.2. Curved weld seam tracking

The curved weld seam is characterized by irregular shapes, and its width and angle may frequently change, which is undoubtedly challenging for seam tracking.

Fan et al. (2017) [66] utilized the support vector machine (SVM) classifier to propose a new seam type identification method, which applies to both straight and curved weld seams. In the computation of input feature vectors, the depth of a series of points of the weld joint was taken as a feature vector. In the model-building part, the SVM-based modeling method and a backpropagation neural network (BPNN) were used to achieve seam type recognition. By experiment on both straight weld seam and curved weld seam, as shown in Fig. 31, this SVM model performed high effectiveness and robustness for all those seam types.

For three-dimensional narrow butt joints, it is hard to detect the weld seam by traditional structured light methods. Zeng et al. (2017) [67] proposed a 3D path guiding method for narrow butt joint welding. A LED surface light source (LSLS) and a cross-line laser light source (CLLS) were used to collect the image of the narrow seam to find the position of the grooves. After the procedure of image processing, as shown in Fig. 32, and information fusion of 2D pixel coordinates and 3D point cloud data, the deviations with an error less than 0.24 mm could be achieved.

4.2.3. Multilayer and multipass weld seam tracking

Multilayer and multipass welding require multiple welding passes to form one or more layers of weld seams. In this kind of welding process, it is often necessary to monitor the welding defects in real-time to distinguish the correct location for every seam, which needs a higher level of real-time performance and precision in seam tracking.

He et al. (2016) [68,69] concentrated on the detection of weld seam profiles and laser stripes for the positioning of welding torches and seam tracking in different layers and passes. They

proposed a model of visual attention guidance, which could simulate the human eyes to enhance the vision saliency of ROIs. In the experiment shown as Fig. 33, this visual attention model successfully emphasized laser stripes, which realized a more accurate seam profile extraction.

Scholars have done much research to solve the problem that small geometrical differences between adjacent weld passes can hardly be detected by traditional structured light. Zeng et al. (2018) [70] developed a method to obtain various kinds of visual information by directional light and structured light. The 3D position of weld passes could be obtained by the composite image of directional light sources (DLS) and structured light sources (SLS), which is shown in Fig. 34. The maximum detection error did not exceed 0.6 mm in experiments.

4.2.4. Application of diverse laser shapes

Single-line laser stripe is the most widely used shape of structured light. A straight laser stripe is projected onto the surface of the workpiece to reflect the characteristics of the weld seam indirectly. In addition to single-line structured light, many other shapes have been studied and applied in active vision weld seam tracking. Those different types of structured light have their characteristics and can have a better performance in different application scenarios [71].

Xu et al. (2008) [72] developed a novel scanning circular laser sensor. The configuration of this vision sensor is shown in Fig. 35. A two-step segmentation (TSS) technique based on partition allocation and local minimum values is presented, and the result of image segmentation is shown in Fig. 36. Then the system calculated three-dimensional coordinates of feature points for welded seam tracking from two dimensions in the image sensitivity plane. The experimental result showed a tracking accuracy of 1 mm in robotic arc welding. Its feasibility of weld seam detection and flatness inspection could be proved.

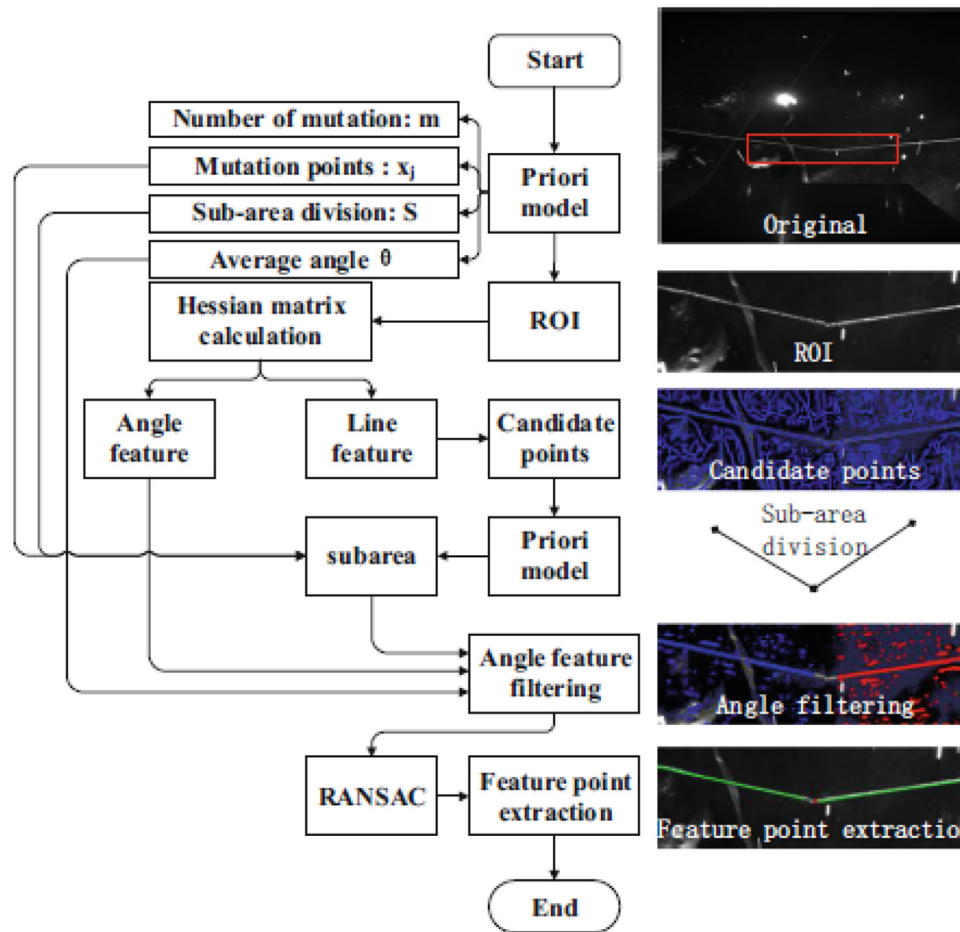


Fig. 30. The flow chart of image processing of improved Steger method [65].

Cross-structured light was used by Kiddee et al. (2016) [73] to avoid complicated computation. A crossed structured light was utilized to set an ROI, and it was detected to locate the feature points by template matching, as shown in Fig.37. An adaptive linear fitting algorithm was employed to estimate the generated template points. According to the experiment, the modified V-groove template performed real-time seam tracking efficiently with high accuracy.

The multi-line structured light consists of several parallel laser stripes, which can provide more information when determining the position of weld seams and has good robustness under interference. Zou et al. (2018) [74] took advantage of the robustness of the three-line structured light in the feature point tracking method. The Gaussian kernel was used to extract feature points in real-time, which is shown in Fig.38. The tracking error was less than 0.32 mm.

Traditional single-line structured light has deficiencies in detecting the seam position and pose in multipass welding. Zhang et al. (2017) [75] presented a new vision sensor system with a grid-like structured laser, which is shown in Fig.39. By extracting feature points from distorted grid shape, the position of seams could be determined after coordinate conversion with an approximation error of 0.5 mm.

4.3. Summary

Weld seam tracking is one of the most critical technologies in automatic robotic welding, which directly affects the final welding efficiency and formation quality. Visual sensing technology is the most commonly used method for seam tracking,

which mainly includes active vision methods and passive vision methods.

The welding process is complicated and changeable. The harsh welding environment, such as strong arc light, large current and smoke, greatly increases the difficulty of weld seam tracking. In addition, seam tracking has high requirements on precision, real-time, and stability. At present, the mainstream of welding seam tracking adopts active vision mainly because the laser structured light is stable, not easy to be disturbed by the external environment, and its processing algorithm is simple. However, the fatal shortcoming of active laser vision is the existence of advance detection and delay within the advance distance. It is also unusable under the partial block of specially shaped workpieces.

Besides, many experts and scholars have conducted researches on weld seam tracking algorithms, and a series of research results have been obtained. These achievements mainly focus on tracking accuracy, real-time performance, and robustness, which are urgent problems to be solved in the weld seam tracking.

5. Weld pool monitoring and penetration control

In automatic welding, the parameters of the weld pool can provide critical information such as width and depth, which directly reflects the welding quality. Therefore, real-time pool monitoring and penetration control are the key steps to ensure accuracy in robotic welding. At present, many approaches are applied to detect the information of molten pool, such as infrared imaging and

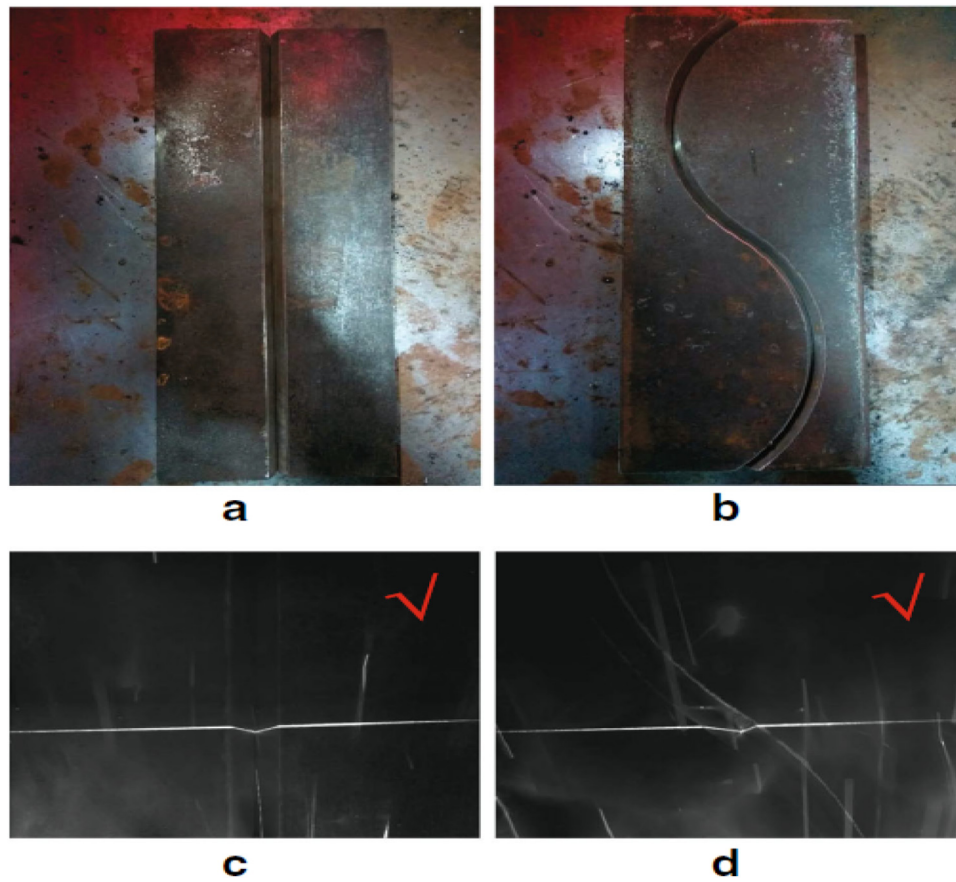


Fig. 31. Testing samples of symmetric V-groove. (a) The straight weld seam with small groove size. (b) The curved weld seam with small groove size. (c) The structured-light image of the straight weld seam with a small groove size. (d) The structured-light image of the curved weld seam with a small groove size [66].

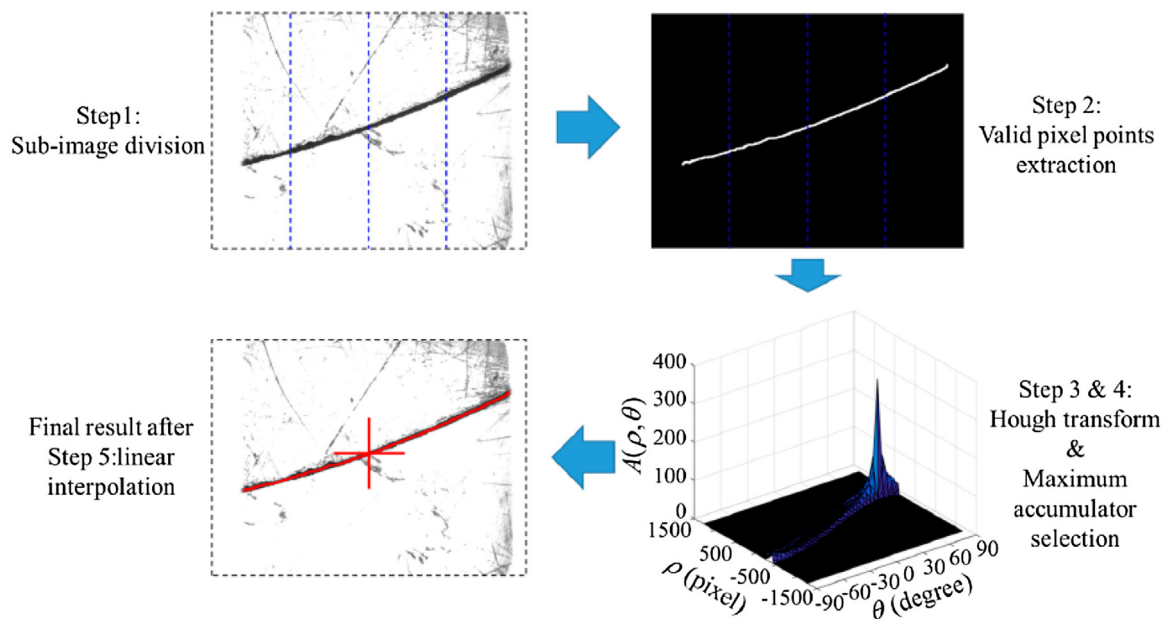


Fig. 32. The output of each step in the proposed narrow butt joint detection algorithm [67].

ultrasonic detection. Because of the straightforward structure and high precision, vision-based monitoring is still the primary choice for weld pool monitoring. Similarly, vision-based pool monitoring methods can be divided into active vision methods and passive vision methods.

5.1. Passive vision-based weld pool monitoring

The passive vision sensors can be used to directly monitor the arc to obtain sufficient and accurate information of the molten pool and observe its dynamic changes in real-time. Besides, the passive

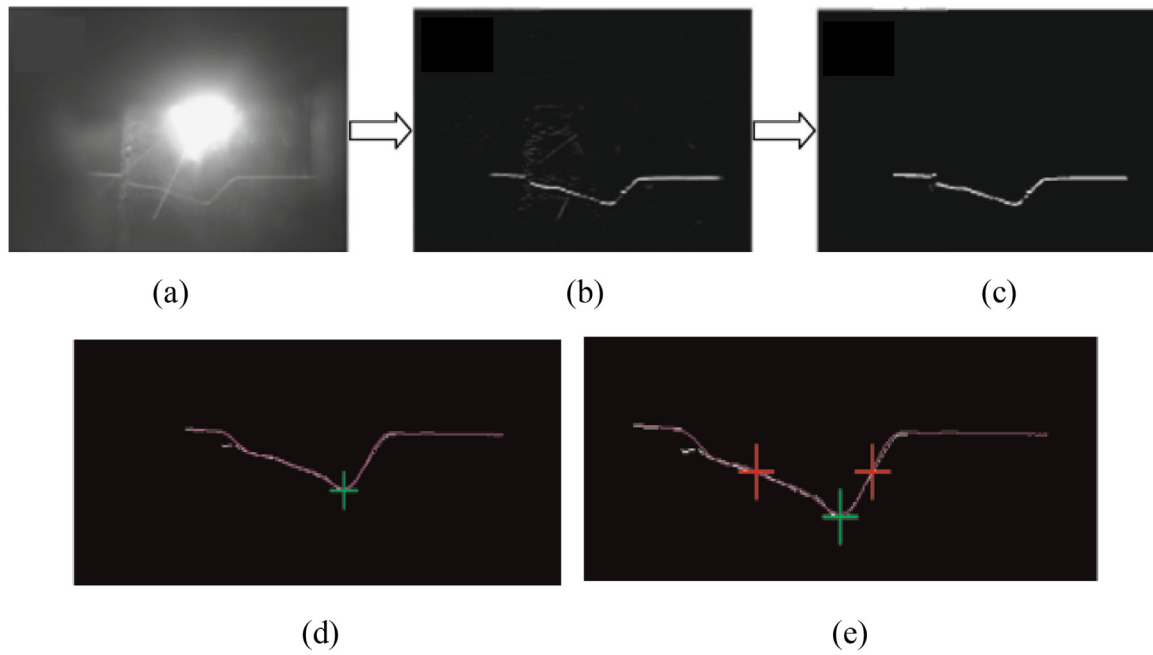


Fig. 33. (a) Image with strong spatter noise. (b) Comprehensive saliency map using the proposed model. (c) Weld seam profile. (d) The extreme point represented by a green reticle. (e) Two red reticles mark the position of inflection points [68].

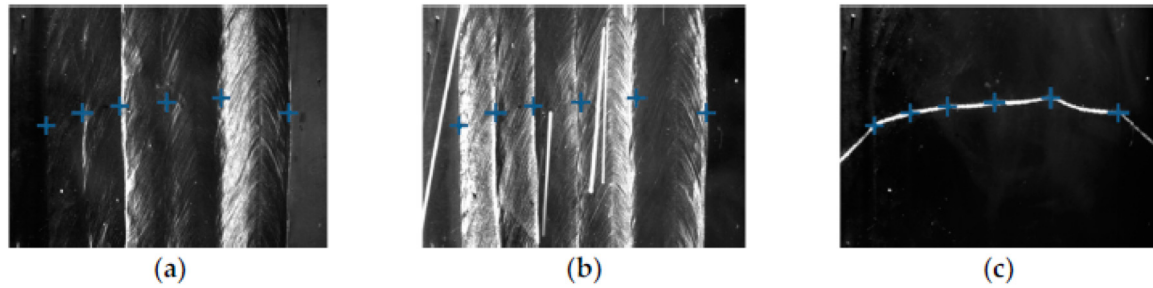


Fig. 34. The detection results marked in different images. (a) Results marked in the image when the left-side DLS is on. (b) Results marked in the image when the right-side DLS is on. (c) Results from composite images [70].

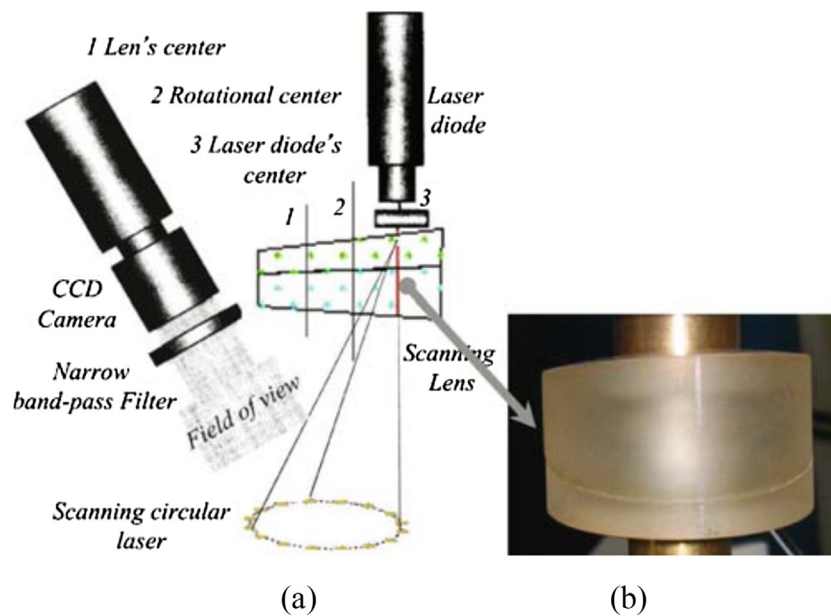


Fig. 35. (a) Equipment of creating a circular laser stripe. (b) scanning lens [72].

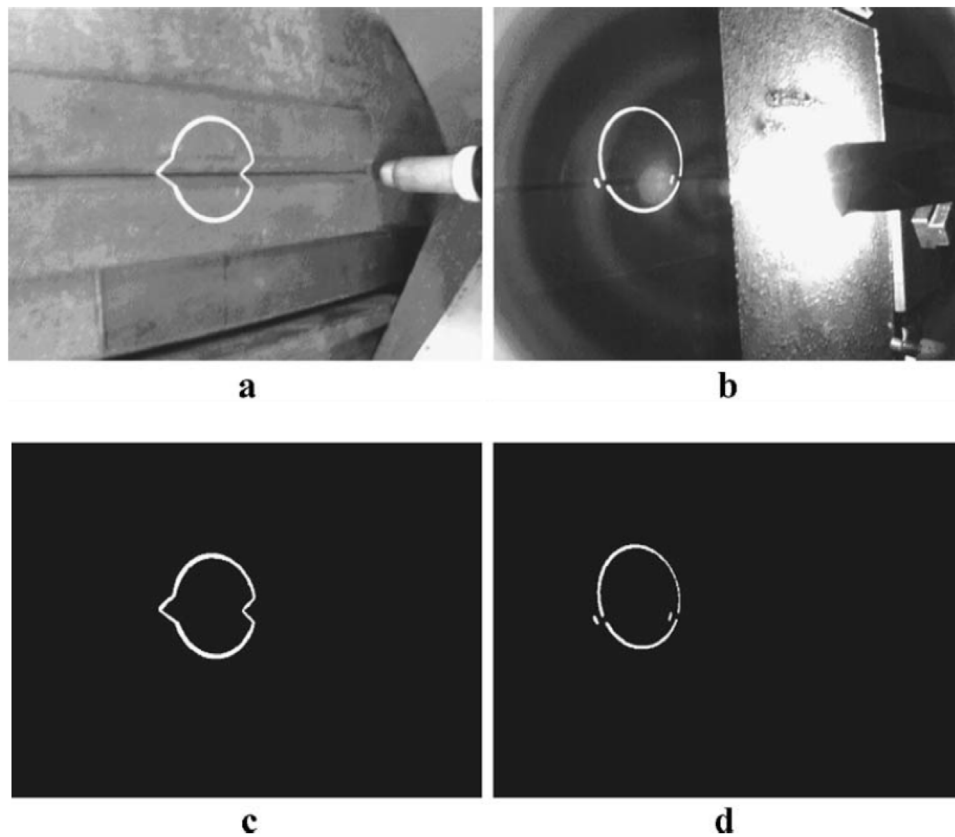


Fig. 36. Results of image segmentation. (a) Original image (V-joint, butt, natural light). (b) Original image (I-joint, butt, gas tungsten arc welding). (c) Image segmentation of image a. (d) Image segmentation of image b [72].

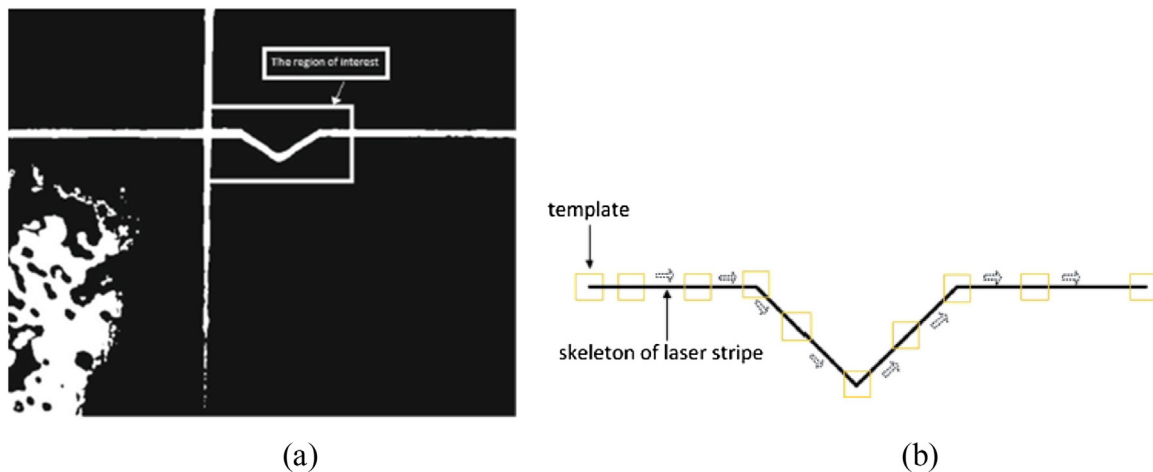


Fig. 37. (a) The region of interest located by cross mark. (b) Illustration of template matching on skeleton points [73].

vision method has the advantage of simple structure and relatively low price. Therefore, it has become the most widely used method in weld pool monitoring and penetration control.

5.1.1. Weld pool monitoring by image processing

Passive vision mainly relies on the direct acquisition of clear images of the weld pool to realize real-time monitoring and control. Therefore, the critical technology of weld pool monitoring is to use various image processing algorithms to accurately extract characteristic parameters such as contour features from welding images.

Wang et al. (2005) [76] designed a new visual sensing system for imaging the topside and backside of the weld pool simultaneously in tungsten inert gas (TIG) welding of aluminum alloy. After the image preprocessing, backpropagation (BP) neural network was used to extract the edge information of the weld pool, as shown in Fig. 40. Because of the symmetry characteristic of the weld pool under large welding current, clear edges of the whole weld pool could be obtained in the unilateral image to adjust the parameters of the weld pool reasonably. The experiment proved that this method could keep the width of the weld seam stable on the aluminum alloy plate with a back width of 6 mm, and could effectively solve the burn-through problem due to the heat sink.

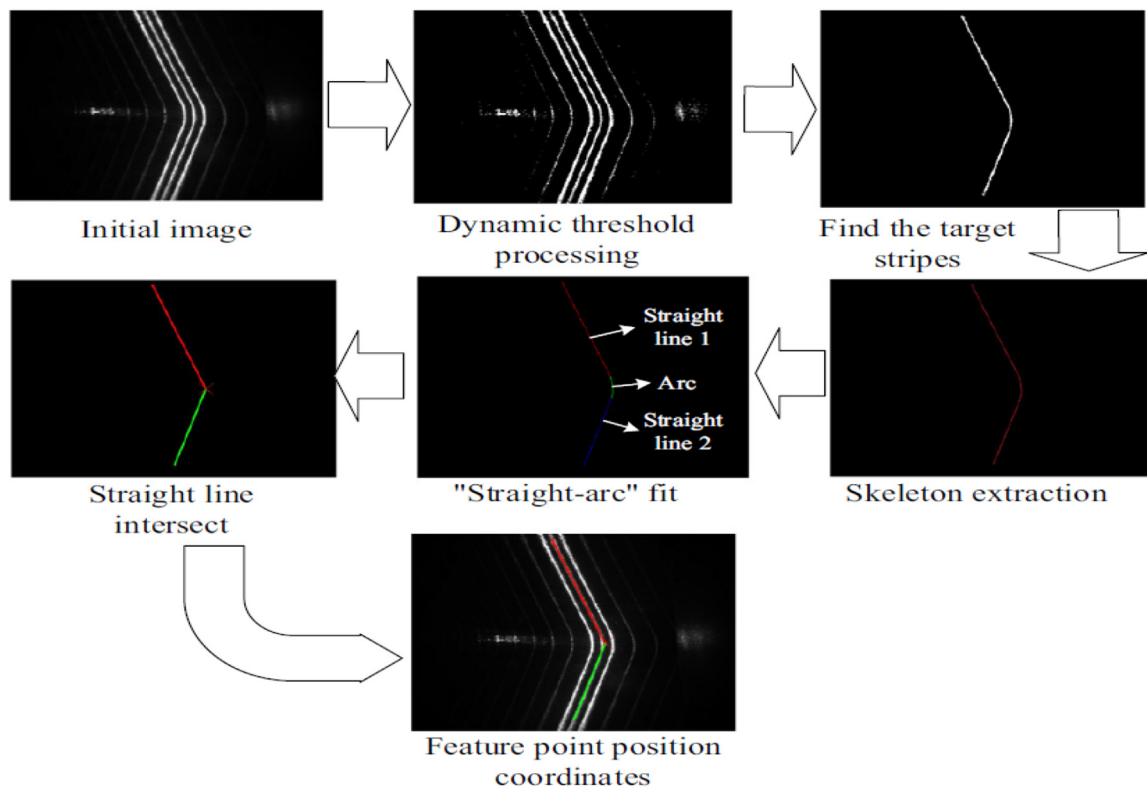


Fig. 38. The extraction process of the initial welding feature points [74].

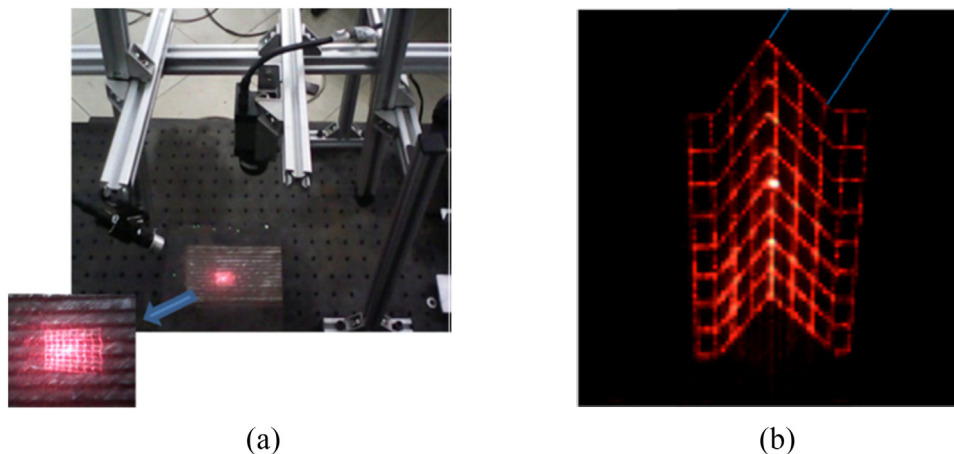


Fig. 39. (a) System of vision sensing platform. (b) Laser grid projected on a V-groove [75].

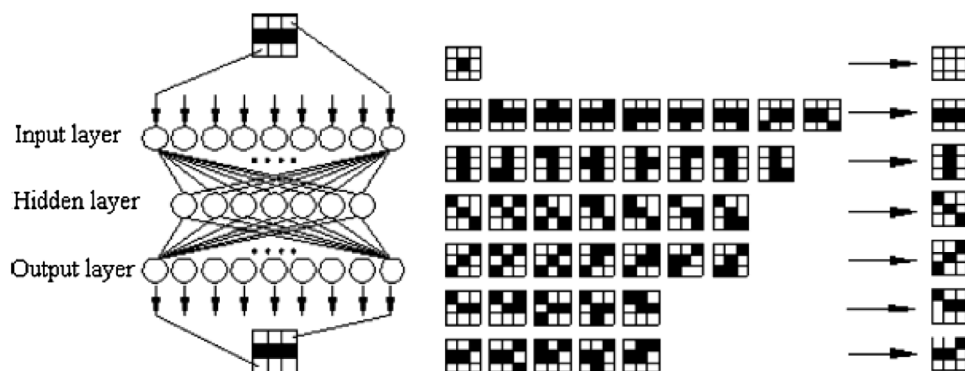
Ma et al. (2010) [77] presented a low-cost binocular vision system that could detect the weld pool geometry and root gap simultaneously. The gray distribution analysis was used to extract the molten pool information, and the back width of the weld pool was predicted in real-time by a neural network. After extracting clear edge information of the root gap and weld pool, as shown in Fig. 41, the system could achieve both weld tracking and penetration control during the welding process as experiments showed on dumbbell-shaped workpieces.

In addition to directly extracting the geometric parameters of the weld pool from the images, heat distribution can also be used to reflect the welding status. Ding et al. (2017) [78] proposed a weld pool monitoring method based on thermal characteristics. The thermal information in the welding process was inferred by gray distribution and the shape of weld pool images, and a

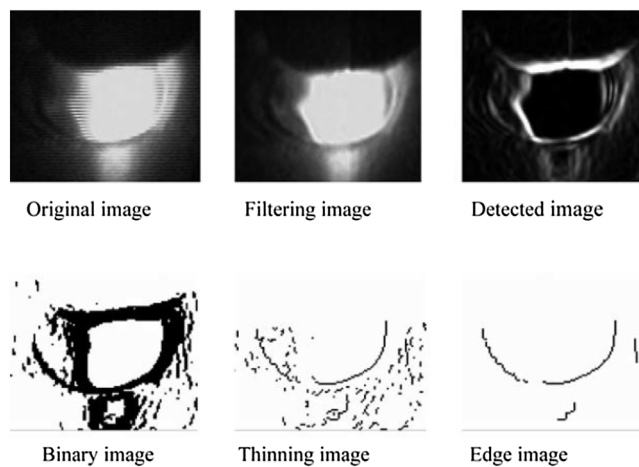
comprehensive prediction model was established. A series of characteristic parameters of the weld pool, such as width, depth, and grayscale centroid, were extracted from the images. The relationship between the heat distribution and the welding condition parameters was estimated by using a neural network with an accuracy of 95 %.

5.1.2. Information extraction from the keyhole

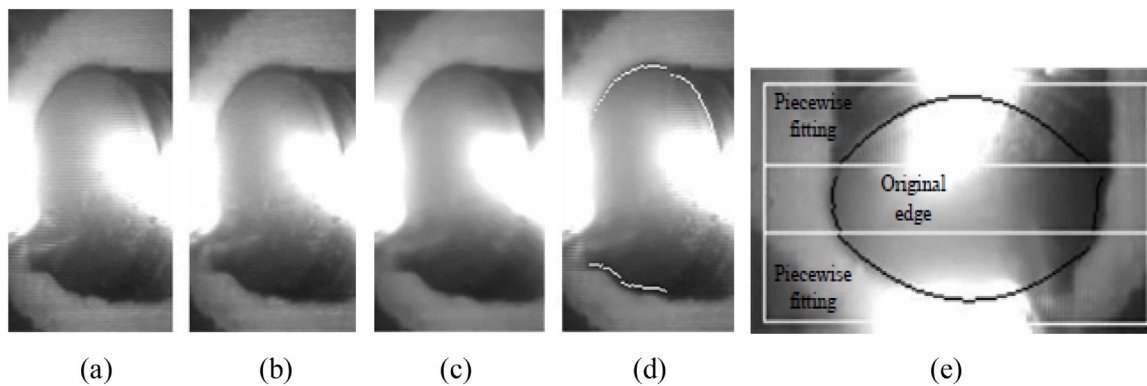
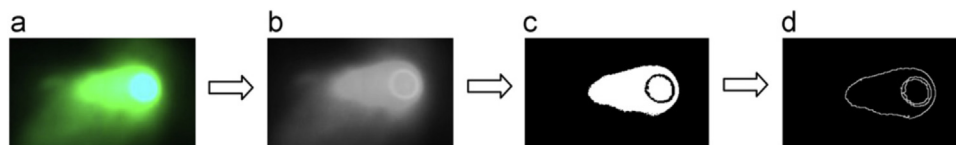
In the welding process, holes generated by melting and gasification when the laser beams project on the workpiece are called keyholes. The shape and depth of keyholes reflect the status and quality of the welding process. Therefore, in addition to monitoring the weld pool, many researchers pay attention to the information extraction of the keyhole.



(a)



(b)

Fig. 40. (a) BP network and sets of the learning patterns. (b) The flow of image processing [76].**Fig. 41.** (a) Original image. (b) Degradation recover. (c) Median filter. (d) Edge search. (e) Contour fitting [77].**Fig. 42.** Image processing procedure: (a) Original image. (b) Greyscale image. (c) Regional growth. (d) Contour detection with the Canny algorithm [79].

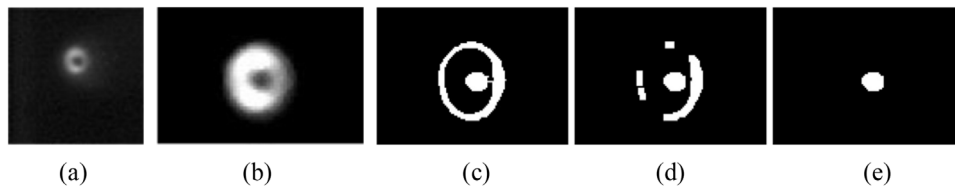


Fig. 43. (a) Original image. (b) Image after enhancement. (c) Region grows operation. (d) Erosion and dilation operation. (e) Image after repeated region grows operations [80].

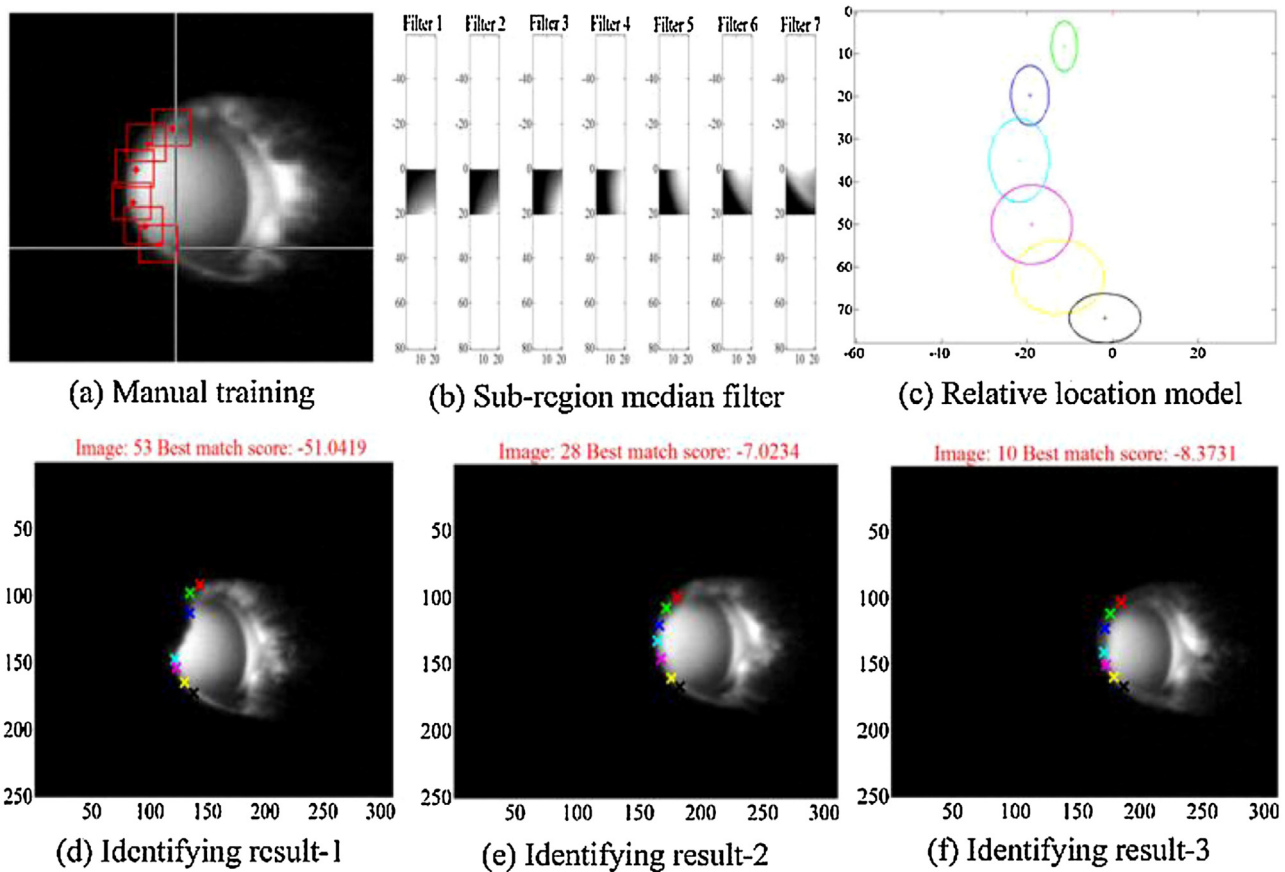


Fig. 44. The localization results of the left-side pool edges [81].

Zhang et al. (2013) [79] established a quality monitoring system for fiber laser welding of galvanized steel. The system conducted high-precision monitoring of weld width and depth by extracting information from the weld pool and pinhole images utilizing region growth and Canny operator. The procedure of image processing is shown in Fig. 42. The welding parameters could be adjusted in time when the weld width was unstable or when the penetration depth was inappropriate, which could avoid possible welding defects.

Zhao et al. (2016) [80] proposed a full-penetration hole detection algorithm based on region growth algorithm, which is shown in Fig. 43. Because the images under different narrow bandwidth filters reflected the temperature distribution of the workpieces, the diameter of the fully penetrating hole could be calculated. By studying the relationship of the diameter of full penetration holes with welding speed and the laser power, this method could identify partial penetration and laser cutting, which was useful in closed-loop controls.

Wu et al. (2017) [81] utilized a vision sensor system to acquire the keyhole image from the back-side of the workpiece in plasma arc welding. The characteristics of keyholes specified by area and tilt angle were extracted by a part-based tree model from the

acquired images, as shown in Fig. 44. A hybrid approach of particle swarm optimization and an adaptive-network-based fuzzy inference system (PSO-ANFIS) was proposed to monitor the penetration status. Experiments showed that this model could well describe the relationship between the characteristics of the keyhole and the penetration status.

In recent years, the application of image enhancement and deep learning also provides new ideas for the monitoring of the molten pool and keyholes. Xia et al. (2020) [82] developed a visual monitoring system for the keyhole TIG welding process. A convolutional neural network (Resnet) was used to recognize different welding states, including good weld, incomplete penetration, burn through, misalignment, and undercut. The training process with a metric learning strategy proved the effectiveness of this keyhole monitoring system.

5.2. Active vision-based weld pool monitoring

At present, some scholars have studied the applications of active vision on weld pool monitoring and penetration control. Although it is difficult to obtain the image of the weld pool by active vision,

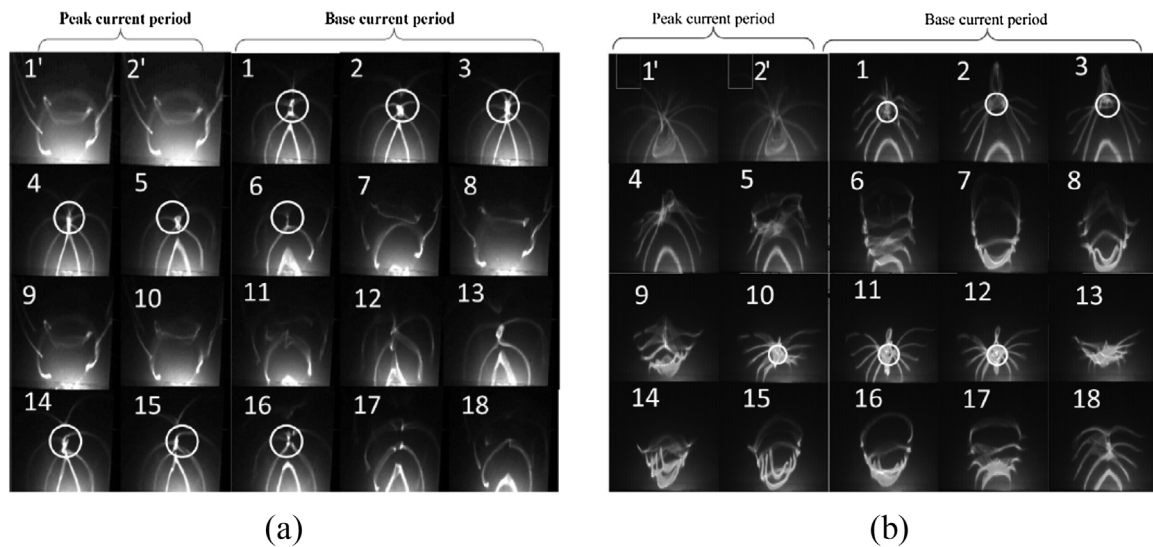


Fig. 45. (a) Typical reflected laser stripe image in full joint penetration. (b) Typical reflected laser stripe image in critical joint penetration [83].

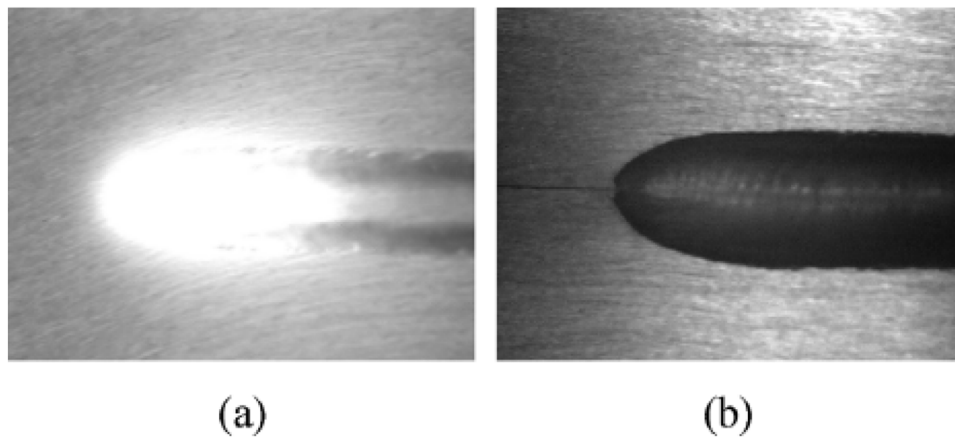


Fig. 46. Influence of auxiliary light source on back-penetration images. (a) Image without auxiliary light sources. (b) Image with auxiliary light source [84].

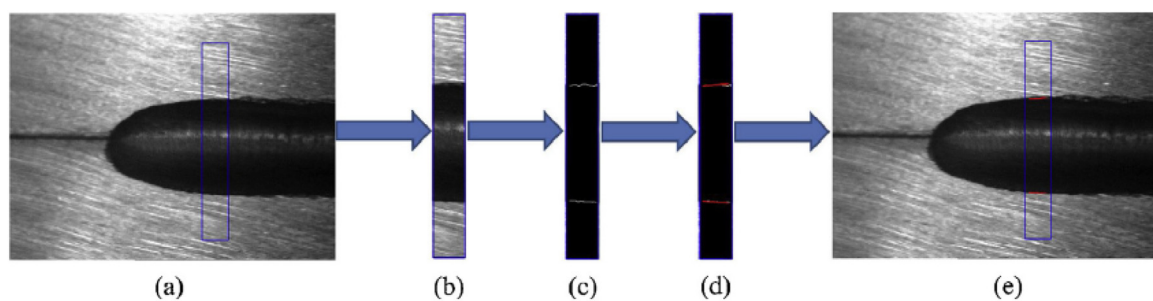


Fig. 47. The procedure of back penetration image processing. (a) Original image. (b) Small window. (c) Edge detection. (d) Features fitting. (e) Image fusion [84].

some characteristic parameters can be obtained indirectly by the image of laser stripes. The laser pattern on the weld pool surface can be used to obtain parameters, such as the length and width, in order to extract the penetration information.

Yu et al. (2016) [83] designed a laser vision sensing system to monitor and study the oscillation behaviors of the weld pool for pulsed gas tungsten arc welding. Periodic changes of laser stripes projected onto the surface of the molten pool could reflect the motion and penetration status, which is shown in Fig. 45. Under the conditions of partial joint penetration and full joint penetration, the pool showed a single oscillation frequency. However, for

critical joint penetration, two oscillation frequencies were generated. Through the relationship between the oscillation frequency and the weld depth, the real-time monitoring of the weld pool can be realized

Xiong et al. (2019) [84] presented a novel active vision sensing system consisting of a camera and an auxiliary light source, as shown in Fig. 46, for back-penetration monitoring in impulse MIG suspension welding of aluminum alloy. The edge detection algorithm was used to detect the back width of the weld pool and reflect penetration status. Under the closed-loop fuzzy controller, uniform back penetration control could complete with a maximum deviation

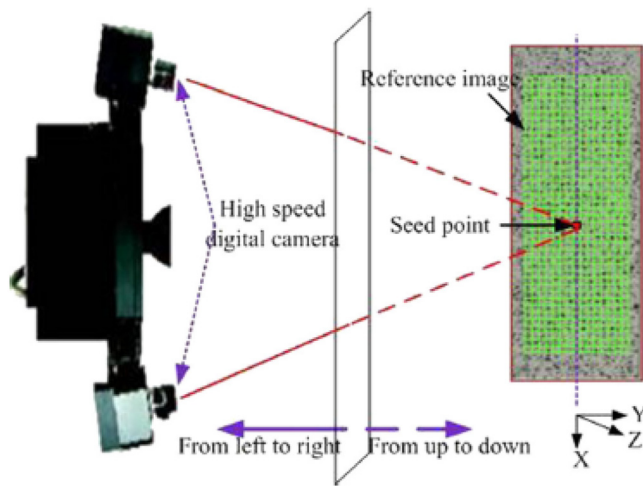


Fig. 48. The schematic of strain field measurement [85].
Chen et al. (2015) [86] proposed a DIC deformation detection

tion within 0.5 mm. The proposed method could also improve weld seam features and anti-interference ability. The procedure of image processing is shown in Fig.47.

5.3. Summary

The main purpose of welding pool monitoring is to monitor the quality of the dynamic welding process, including the state of the front pool and the state of the back penetration. At present, passive vision is widely used to solve this key problem for intelligent quality control of robotic welding. A qualified weld seam should have not only good forming quality but also good internal quality, which is highly dependent on the monitoring of the welding pool.

Weld pool monitoring is extremely difficult, mainly because the external welding environment is very harsh. The serious interference affects the image acquisition of the weld pool and puts forward higher requirements for the image processing algorithms. Current research mainly focuses on how to collect clear images of the weld pool, develop powerful image processing algorithms, improve the welding quality prediction, and analyze welding defects.

6. Digital image correlation method for deformation measurement

In the process of robotic welding, the workpiece is inevitably deformed due to heat accumulation. The deformation will directly affect the shape and structure of the weld seam, so real-time deformation detection is critical to improving the welding quality. Due to the high temperature and strong arc interference in the welding process, the applications of traditional contact deformation measurement is restricted. With the continuous development of visual sensing technology, the Digital Image Correlation (DIC) method of measuring the surface displacement field by comparing images in different states has been widely used in deformation detection.

6.1. Digital image correlation methods

Li et al. (2014) [85] utilized 3D digital imaging technology and inverse modeling methods to determine the mechanical properties of weld seams in the heated process. Deformation hardening exponent and strength coefficient in different welding areas were determined by DIC technique and inverse modeling, respectively. The indentation test showed that the method is effective in identifying mechanical properties. The schematic is shown in Fig.48.

technology aiming at the defect that traditional DIC methods are easily affected by high temperature and intense arc light. This method utilized a unique surface preparation method to produce speckle patterns which could sustain high temperature. An optical illumination and a filter system were applied to suppress the influence of intense arc light. The thermodynamic finite element model experiment in Fig.49 showed that the technology enables strain field measurement at a distance of 1 mm.

The traditional two-dimensional DIC methods only apply to the deformation measurement in flat surfaces. A three-dimensional DIC deformation detection method was proposed by Chen et al. (2018) [87] to measure the out-of-plane deformation in the welding process. An auxiliary diffusive speckle pattern was used to assist subset matching, as well as the determination of the initial 3D global coordinate of each subset when the object is undeformed. The process is shown in Fig.50. The feasibility of the proposed method was proved by two experiments of out-of-plane rigid body translation and uniaxial tensile testing.

Chen et al. (2016) [88] designed a multi-sensor system for in-situ deformation detection based on DIC and infrared radiation (IR). Temperature and strain near the weld seams can be measured after the DIC full-field noise reduction by appropriate filters. Finally, the system could reflect the changes in welding strain and temperature fields under different conditions. It associated those parameters with the weld attribute to guide the robotic welding process.

6.2. Summary

Digital image correlation (DIC) is a widely used method for deformation measurement in the welding process. Compared with the limited traditional strain detection methods, the DIC methods have more application scenarios, which is especially suitable for real-time online measurement. In recent years, researchers focus on new technologies such as 3D detection and multi-sensor information fusion, which improve the adaptive capacity of traditional DIC methods.

7. Defects detection and quality inspection

Welding quality inspection and defect inspection are critical steps in the automatic welding process. However, traditional defect inspection methods based on personal experience is not always reliable. Thus, vision-based defect detection methods are widely used to inspect the uneven surface, press marks, and other possible defects in the welding process. It can also monitor the width, height, and other parameters of the weld seams in real-time [89]. According to different methods of image acquisition and feature extraction, visual-based defect detection methods are divided into passive vision, active vision, and radiographic inspection.

7.1. Passive vision defects detection

The passive vision-based defect detection method mainly uses CCD cameras to directly obtain images of the weld area, which is similar to human inspection. After obtaining the images, the difference between the target weld seam and the standard weld seam is often identified through a comparison of welding parameters and image matching. Therefore, the defects in the welding process can be determined.

To overcoming certain limitations such as the inaccuracy, non-uniformed illumination, and deficient contrast, Kumar et al. (2012) [90] designed a new visual inspection system for metal inert gas (MIG) welding. From the obtained images, the regions of interest were segmented, and the average gray level of the characteristic features was calculated. The quality of weld joints was classified

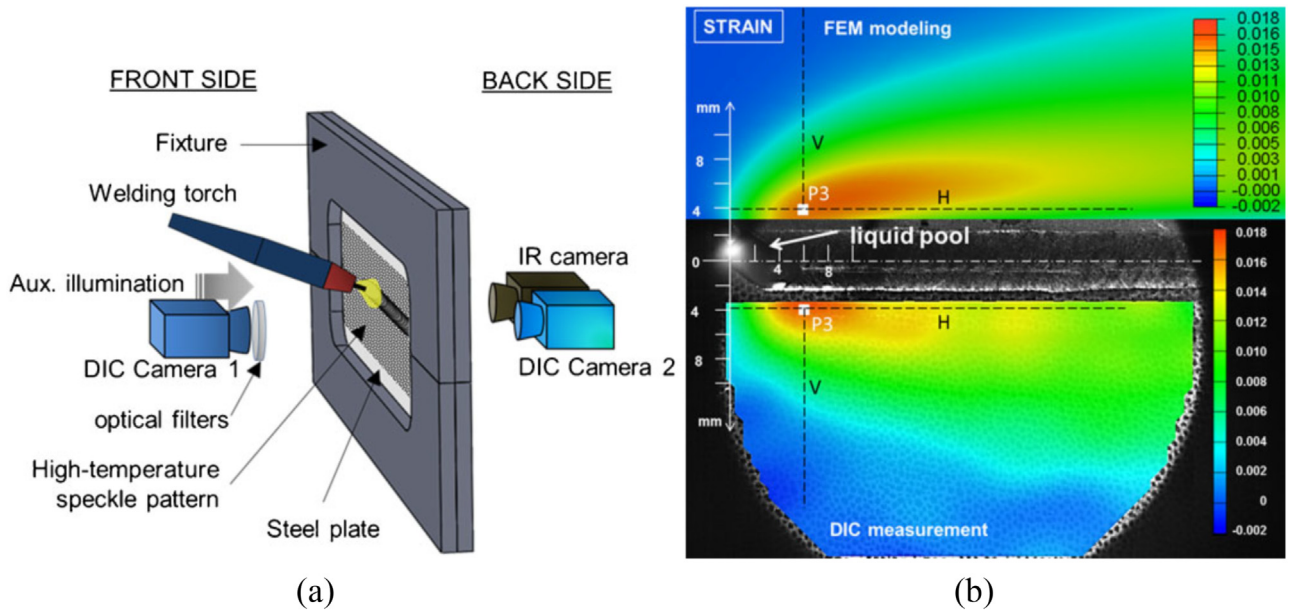


Fig. 49. (a) The schematic of the experimental set-up. (b) The comparison of transverse strain field and experimental measurements by new DIC technique [86].

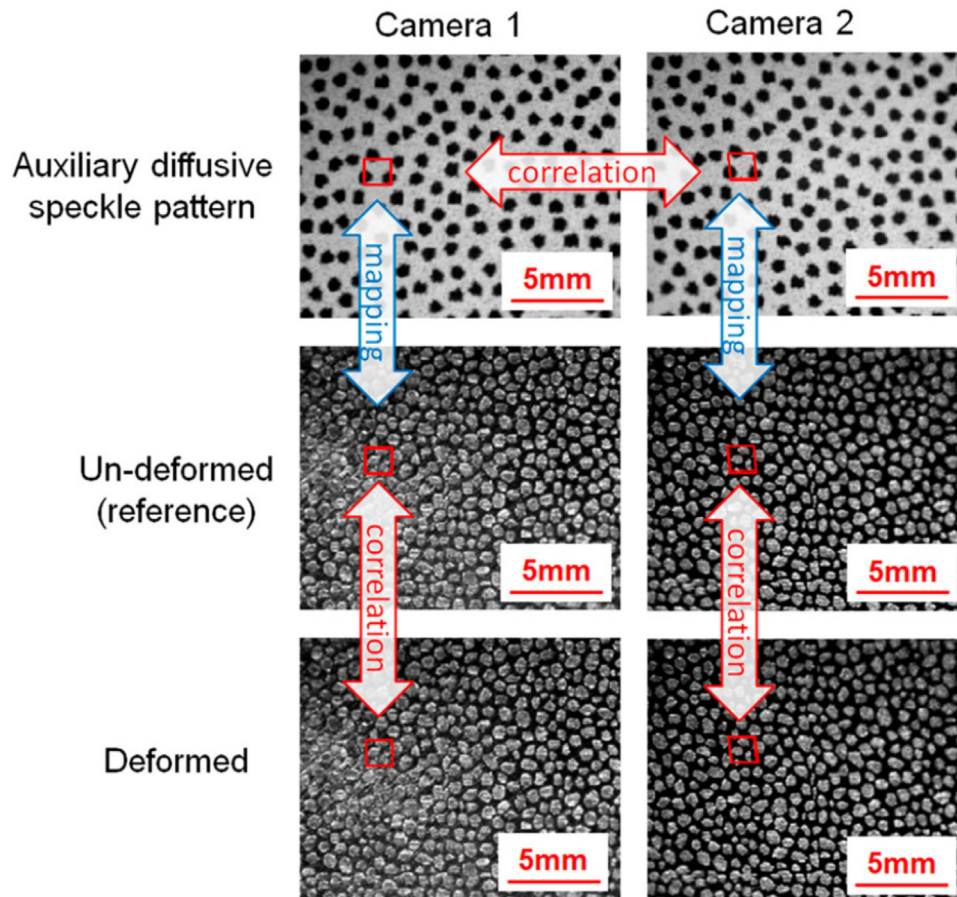


Fig. 50. The 3D correlation strategy on the specular speckled surface with the assistance of an auxiliary diffusive speckle pattern [87].

into four predefined types by the back-propagation neural network with an overall accuracy of 95 %, as shown in Fig. 51.

For multipass thick steel plate welding, He et al. (2019) [91] modeled the typical welding process by a finite state machine (FSM) for process monitoring and fault correction. If the parameters in

the model are not appropriate, the system will terminate the welding process. A new contour extraction algorithm was proposed to change the experience parameters adaptively. The experimental results showed that this method has a 95 % contour extraction rate and a high fault detection rate.

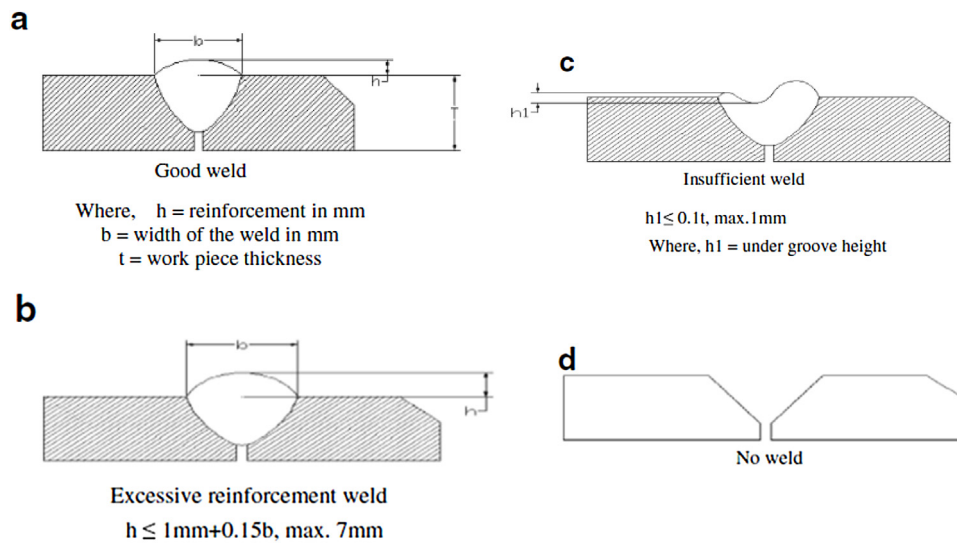


Fig. 51. Different types of acceptable and unacceptable groove weld profiles in the butt joint. (a) Good weld. (b) Excessive reinforcement weld. (c) Insufficient weld. (d) No weld [90].

7.2. Active vision defects detection

As discussed earlier, the active vision-based welding system can obtain a series of welding parameters by locating feature points of laser stripes, so it can also be used for quality inspection. Three-dimensional weld seam reconstructions are usually applied to determine the status in the welding process and to locate the weld defects in real-time.

Li et al. (2010) [92] designed a structured light-based vision inspection system for weld bead profile measurement, monitoring, and defect detection. By the image processing algorithms for laser profiles and feature points, the dimensional parameters of the multilayer weld bead could be measured. The quality of the weld seam could be inspected by the reconstructed profiles, as shown in Fig. 52.

Nguyen et al. (2014) [93] developed a quality inspection system based on the principle of laser triangulation. A sliding vector method was used for the feature point extraction, and the 3D weld surface reconstruction can be completed. By comparing the standard dimensions with the dimensions obtained through the measurement procedure, the defects could be determined, as shown in Fig. 53.

7.3. Radiographic defects detection

Radiographic inspection uses the photosensitive properties of X-ray and determines whether there are defects inside the work-pieces by observing the attenuation changes on the recording film. In practice, the radiographic inspection is often used for non-destructive examination.

Hassan et al. (2012) [94] presented a novel technique for the detection and classification of weld defects by geometric features. Defects in denoised images were localized with maximum interclass variance and minimum intraclass variance. Finally, the geometric descriptors in segmented images were used for defects classification by an Artificial Neural Network (ANN). The accuracy of the proposed method was found to be more than 91 % in defects detection and 96 % in defects classification.

Besides the geometric features, Valavanis et al. (2010) [95] also utilized texture features for detecting and discriminating discontinuities in weld images. A set of descriptors corresponding to texture measurement and geometrical features were extracted for each

segmented object and given to a classifier as inputs, as shown in Fig. 54. Three different classifiers (Support Vector Machine, Neural Network, k-NN) were tested by cross-validation, and the possibility of reducing computational cost was also be studied.

7.4. Summary

Weld defect detection and quality inspection are highly dependent on visual sensing technologies. At present, both passive vision and active vision are used. The research mainly focuses on the automatic recognition of weld surface quality by using visual sensing technologies instead of manual operation, as well as the further study of defect recognition and classification according to the images.

8. The future interests of vision-based technology in robotic welding

In recent years, there have been many developments in the research and applications of visual sensing technologies in robotic welding, including calibration, guidance, weld seam tracking, weld pool monitoring, penetration control, and welding deformation detection. Many research results have been obtained, but improvements are still needed. The key research direction of vision-based technology in robotic welding may include the following aspects.

8.1. Hardware developments of vision sensors

The breakthroughs in hardware technologies of charge-coupled device (CCD) and complementary metal-oxide-semiconductor (CMOS) promotes the development of visual sensing technologies in the robotic welding, such as high dynamic range CMOS and high-speed photography.

As welding processes are accompanied by strong arc light, CMOS cameras with high dynamic range can be used to collect welding images of higher quality, which reduces the difficulty of image processing and makes welding control much easier. Nicolosi et al. (2012) [96] proposed a spatter detection algorithm based on a typical cellular neural network using a CMOS camera with a wide dynamic range. Leonardo et al. (2016) [97] used a single CMOS camera as the vision sensor and proposed a groove measurement system by Vision-Based Measurement (VBM).

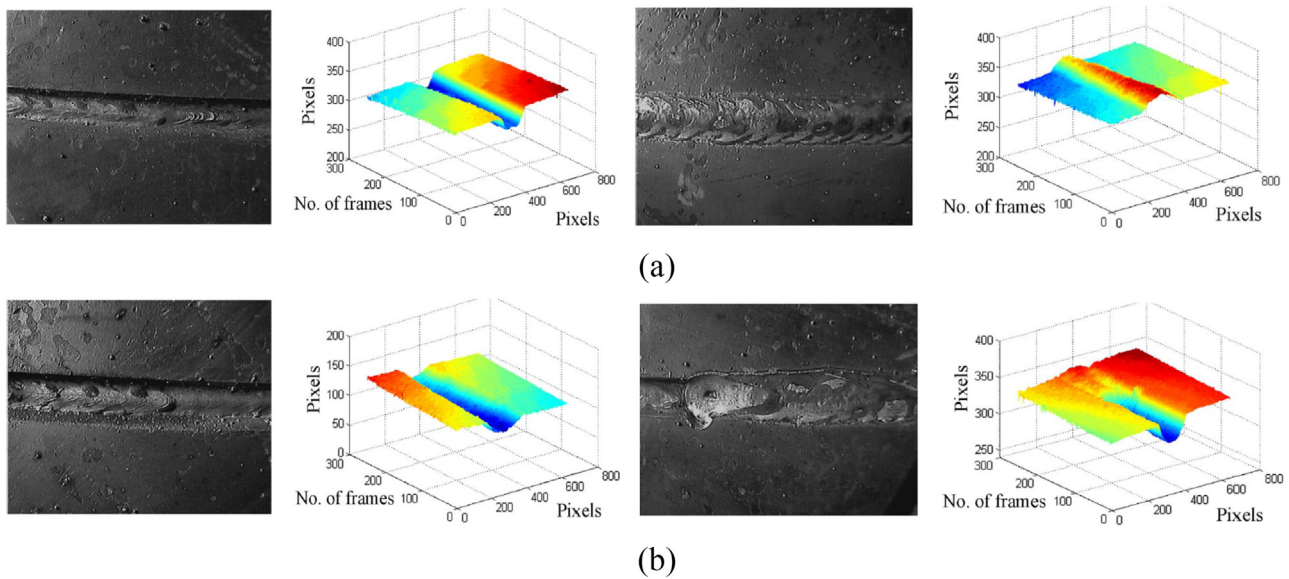


Fig. 52. (a) Photos and profiles of the weld beads in defect-free multilayer weld. (b) Photos and profiles of the weld beads in multilayer weld with defects [92].

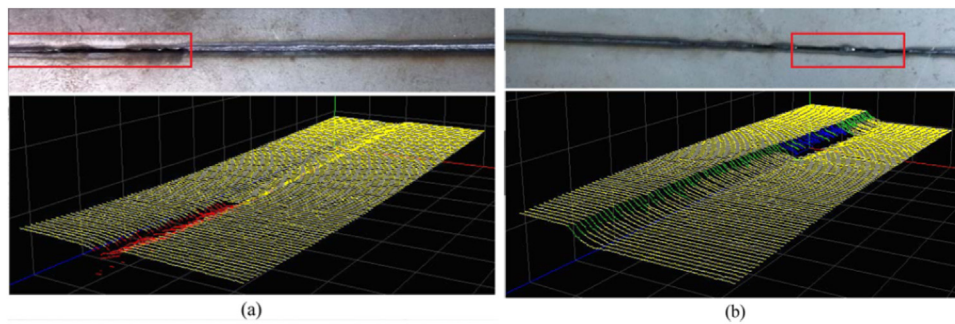


Fig. 53. 3D weld surfaces with detected weld defects. (a) Pinhole error. (b) Pinhole error and mismatch error [93].

In order to monitor the plume and splash during the laser welding process, You et al. (2014) [98] presented a highspeed photography system combining ultraviolet sensing and visible sensing, as shown in Fig.55. The future development of visual sensing hardware will greatly promote the application of visual sensing technology in intelligent robotic welding.

8.2. Composite vision of passive vision and active vision

The visual sensing technology combining active vision and passive vision becomes a new trend in robotic welding applications. It combines the advantages and makes up the defects of those two visual sensing methods.

Xu et al. (2016) [99] analyzed the advantages and drawbacks of both active vision and passive vision and accordingly designed an improved vision sensor. The proposed sensor has two CCD cameras for active vision and passive vision separately, which is shown in Fig.56. The composite sensor can combine the advantages of two visual sensing methods to get more information from the weld seam and weld pool. However, the research of the composite sensor is still at an early stage, which needs to be further studied in the future.

8.3. Advanced image processing algorithm

The quality of image processing algorithms directly affects to the precision of the initial position finding, seam tracking, and other

processes in automatic welding. Image processing algorithms with fast speed, simple process, high precision, and strong robustness are urgently required in noisy welding environments [100]. The optimization of image processing algorithms can be mainly considered from the following aspects, including accuracy, real-time performance, and robustness.

In order to realize autonomous welding, it is necessary to guide the robot to the initial welding position accurately, then conduct precise weld seam identification and real-time weld seam tracking. So the image processing algorithms need to be very accurate, especially for the automatic welding of the metal sheet. A little deviation may lead to failure in the welding process, so it is particularly important to develop high precision recognition and feature extraction algorithms. Recently, a variety of sophisticated edge detection operators have been widely used, which can accurately extract the contour of images in different welding scenarios. An improved Canny edge detection method was proposed by Xu et al. (2012) [101]. This new algorithm is devoted to solving the problem that traditional methods cannot remove environmental noise and require manual thresholds setting. Experiments showed that the improved method is superior to traditional Canny operators in both accuracy and robustness.

The real-time performance of image processing algorithms is also significant because of the dynamically changed status of weld seams and weld pools in welding processes. The robotic welding system must be able to respond to those changes in time. There are mainly two methods to improve the real-time performance of

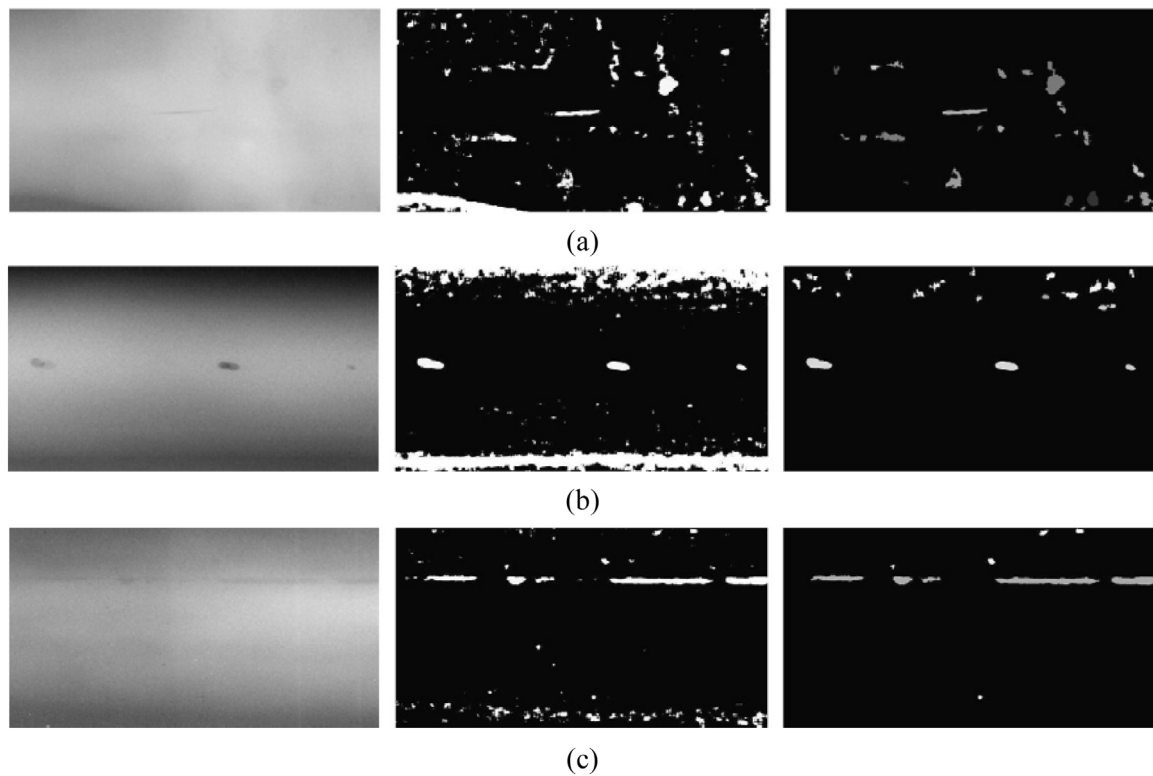


Fig. 54. Original, thresholded and segmented images for welds that contain: (a) Crack (linear in the middle). (b) Gas pores (ovals in the middle). (c) Lack of fusion (linear at the top) [95].

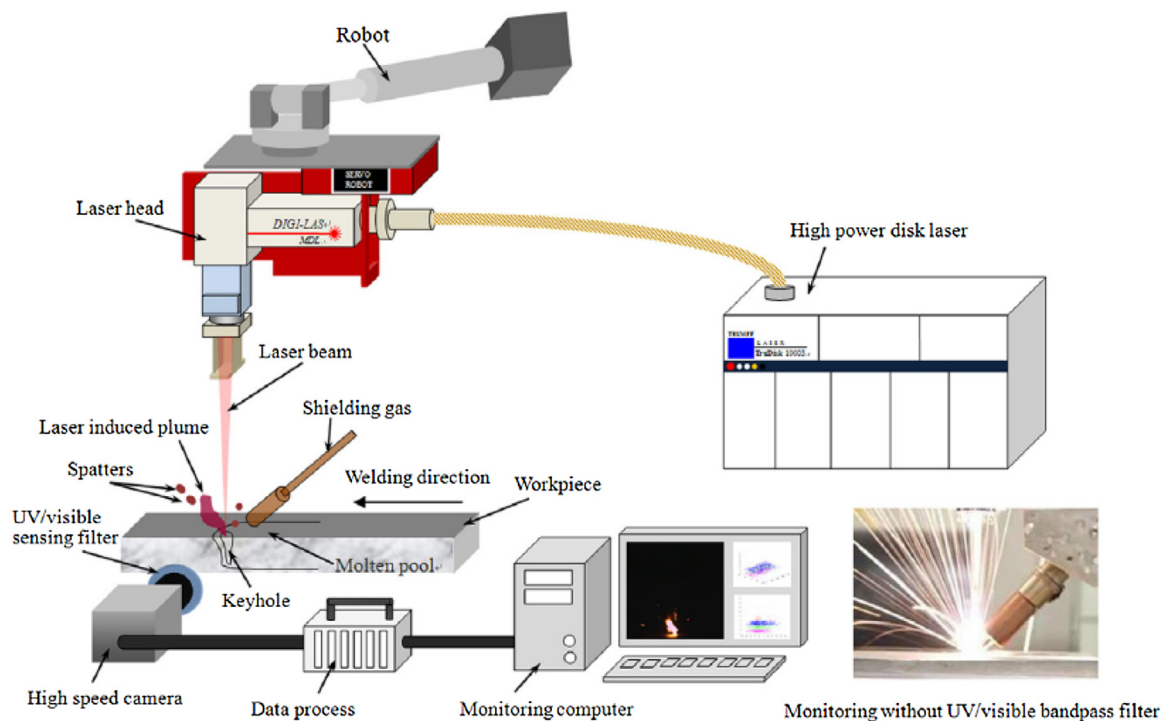


Fig. 55. The monitoring system of high-power disk laser welding based on highspeed photography technologies [98].

robotic welding. One is to adopt more advanced hardware with better performance, which will undoubtedly greatly increase the cost and further limit the popularization of robots due to the higher price. Another is to improve and optimize the welding image processing algorithms, which is the mainstream practice at present. For

example, He et al. (2016) [72] adopted a visual attention model to simulate the selecting mechanism of the human visual systems and obtain relevant information from complex scenes. By locating the focus of attention in a small region of interest (ROI) and excluding

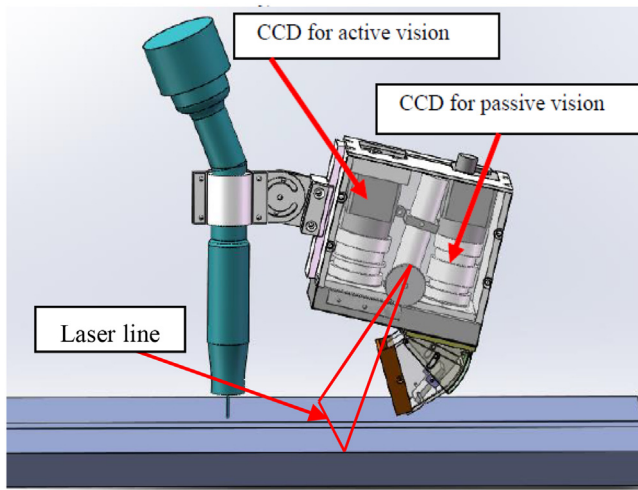


Fig. 56. The composite vision of passive vision and active vision [99].

other invalid background information, the efficiency of subsequent image processing can be significantly improved.

Different from many applications of vision sensing technology, robotic welding is a complex and changeable process, and robustness is a key point that must be solved. Robustness is the guarantee that welding robots can work stably and efficiently for a long time. Due to the complexity of welding environments, image processing algorithms must be able to achieve satisfactory results under possible noise interference, such as splash and smoke. Du et al. (2019) [102] utilized fast image segmentation, convolutional neural network (CNN), and the feature search technique to identify the weld seam features accurately under arc noise. Automatic threshold selection reduced the difficulty of parameter adjustment and increased the stability of seam tracking. This method performs well in processing welding images with significant interference.

8.4. Multi-sensor technology and information fusion

Due to the complexity and variability of the welding process, many factors affect the final welding quality. In order to improve the stability, precision, and forming quality of intelligent robotic welding, information fusion based on multi-sensor technology is the future development trend.

Vision sensing technology is critical to realize intelligent welding control online. However, a single type of vision sensor has some shortcomings in information comprehensiveness and reliability. In the case of severe interference such as arc light, feature extraction is difficult, and image processing algorithms can be very complicated. If multiple sensors are deployed in the robotic welding process to detect the context information online, the temperature, electrical signal, images, and other multi-source information can be obtained and fused synchronously. Therefore, the redundancy and complementarity of different sensors can be fully utilized through the information fusion process. This method can provide a more comprehensive, reliable, and accurate description of the dynamic robotic welding process, which is the main research direction in the future. For example, Hu et al. (2012) [103] proposed a weld seam tracking algorithm based on multi-sensor information fusion, which utilized both visual sensors and arc sensors. Combining the redundant information and complementary information obtained by two sensors under the feature layer can realize deviation correction with improved accuracy of 0.4 mm. Besides, Liu et al. (2019) [104] established a multi-sensor fusion system, as shown in Fig. 57, to predict the real-time status of high-power laser welding.

8.5. Application of artificial intelligence technology

Artificial intelligence, with the abilities of learning, reasoning, and self-correction, is the most popular field of intelligent robotic welding. In recent years, machine learning algorithms are widely used in robotic welding, which provides new ideas for solving traditional welding problems. The comprehensive learning can significantly improve the accuracy and efficiency by acquired data without human intervention or assistance. Because of the powerful ability to process sophisticated features in weld seam recognition, penetration control, and defects analysis under complicated conditions, machine learning is well suited to information-rich visual images.

For example, Du et al. (2019) [21] applied a purpose-built visual sensing system, fast image segmentation, feature area recognition of the Convolutional Neural Network (CNN), and a feature search technique to identify the weld seam features accurately. In addition, robot welding, researchers try to adopt different kinds of machine learning methods on the penetration prediction, such as the Convolutional Neural Network (CNN), Back-Propagation Neural Network (BPNN), Extreme Learning Machine (ELM). Besides, defect identification also benefits from the powerful feature extraction ability of the machine learning algorithms. After acquiring a large amount of data, the robot can make empirical judgments to recognize various defect types. Zhang et al. (2019) [105] studied deep learning-based online defects detection for aluminum alloy in robotic arc welding using CNN and weld images. Yang et al. (2020) [106] proposed a novel weld bead detection and location algorithm based on deep convolutional neural networks. The deep semantic segmentation network could effectively satisfy the robustness and precision requirements for weld bead detection and location.

With the development of artificial intelligence technology, it will be more frequently applied in vision-based intelligent robotic welding. Machine learning algorithms provide reliable precision for automated robotic welding and improve its stability in a variety of complex environments. Especially, CNN can utilize more comprehensive welding features while balancing processing speed and precision to meet the high real-time requirements of automatic welding.

8.6. Summary

- 1 The hardware developments of vision sensors are the basis of vision-based technology. Developed from the traditional CCD and CMOS cameras, the new technologies of high dynamic range CMOS and high-speed photography provide more efficient approaches to collect visual information in welding processes.
- 2 Active vision-based methods and passive vision-based methods are two mainstream visual technology applied in robotic welding. The composite vision is a newly developing trend that combines the advantages of both active vision and passive vision.
- 3 The visual robotic welding technologies based on image acquisition strongly depend on the advanced image processing methods. In order to meet the special requirements of welding tasks, future image processing technology will continuously improve their accuracy, real-time performance, and robustness.
- 4 In addition to the visual sensing welding systems, other sensing technologies can provide the necessary information in the welding process from different perspectives. In the future, the multi-sensor welding system and data integration technologies will become an important research direction of automatic welding.
- 5 Artificial intelligence, with the abilities of learning, reasoning, and self-correction, is the most popular field of intelligent robotic welding. For example, machine learning algorithms are often used in seam recognition under complex conditions, real-time

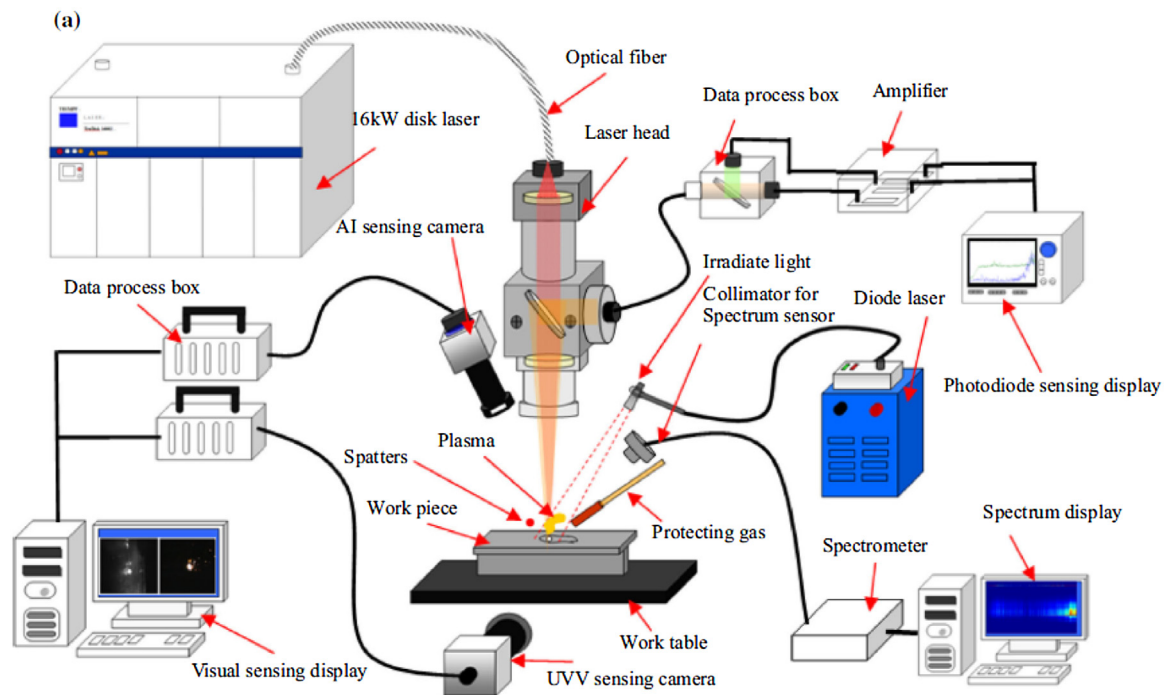


Fig. 57. The experimental setup of high-power laser welding with multi-sensors [104].

seam tracking, penetration control, and welding defects classification. With the development of relevant research, artificial intelligence technologies based on vision sensing will be increasingly applied in intelligent robotic welding.

9. Conclusion

Vision-based technologies play essential roles in robotic welding because of its significant advantages compared with other sensing methods, which attract increasing attention of scholars. This paper summarizes the research developments of robotic welding based on vision sensing technologies in recent years, including visual calibration, initial welding position guidance, weld seam tracking, weld pool monitoring, correlation strain detection, and defects inspection. With the improvement of image processing algorithms and the application of machine learning, the vision-based robotic welding system has made great progress in precision, adaptability, and real-time performance.

Although today's vision-based welding technology still needs to be improved in reliability, this paper can prospect the promising development direction of robotic welding technologies, including the development of visual hardware, optimization of image processing algorithm, the composite vision of passive vision and active vision, multi-sensor information fusion, and applications of machine learning. It can be predicted that with the development of vision sensing technology, robotic welding will gradually reduce the dependence on human operations, and intelligent robotic welding will finally be realized.

Declaration of Competing Interest

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

Acknowledgments

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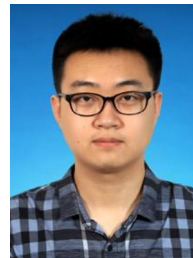
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